

AquaCrisis: A Multilingual Benchmark for Classifying Water-Utility Public Communication

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Abstract

Water utilities communicate both routine service information and time-sensitive public-risk guidance, yet multilingual NLP benchmarks for institution-authored infrastructure communication remain limited. We introduce AquaCrisis, a corpus of 20,000 text-bearing Facebook posts from 12 European water utilities, with a 1,500-post manually annotated gold standard. We formulate Task A as classification of a post’s main water-utility topic and Task B as classification of its primary communicative function, together with a coarser grouped Task B.1. We compare rule-based and supervised classifiers with four instruction-tuned LLMs on native-language and English-translated inputs. Supervised approaches achieve the highest point estimates on the fine-grained tasks: translated TF-IDF with logistic regression performs best on Task A, while translated multilingual sentence embeddings perform best on Task B. GPT-4o-mini achieves the highest point estimate on grouped Task B.1. Paired bootstrap analysis shows that translation effects are model- and setting-dependent: gains are clearest for the lexical TF-IDF classifier and selected Gemma 4 and GPT-4.1-mini conditions, while multilingual sentence embeddings, GPT-4o-mini, and Qwen 3.5 are comparatively stable across input representations. AquaCrisis provides a benchmark for multilingual water-utility communication spanning routine, educational, operational, and incident-related messaging, supporting communication monitoring, operational triage, and cross-lingual evaluation.

1 Introduction

Water utilities are frontline actors in climate adaptation because they connect climate-driven hydrological change with public protective action (Boholm and Prutzer, 2017). Climate change is intensifying water scarcity, drought, rainfall extremes, flooding, salinity, contamination risk, and pressure on urban water infrastructure (Boholm and Prutzer, 2017;

Bradford et al., 2017). At the same time, utilities increasingly use social media, including Facebook, to communicate routine updates, water-quality information, public engagement, and emergency messages (Sallaku et al., 2017; Martin, 2017; Mix et al., 2020). Importantly, incident-related messages occur within a much larger stream of routine, educational, customer-service, and community-facing communication.

This communication matters because failures in water-risk messaging can have direct public-health and adaptation consequences (Day et al., 2024; Bradford et al., 2017; Rundblad et al., 2010). Studies of “Do Not Drink” and “Boil Water” notices show that unclear messages can lead to confusion and non-compliance, while effective communication helps the public take protective action during contamination events, outages, and compound disasters (Day et al., 2024; Rundblad et al., 2010). In a warming climate, timely and understandable communication about contamination, disruption, conservation, and water stress is therefore part of adaptation capacity, not only public relations (Boholm and Prutzer, 2017; Mix et al., 2020; Day et al., 2024; Bradford et al., 2017).

Water-utility communication both describes an issue affecting the water system and performs a communicative action toward the public. We therefore formulate the problem as two complementary tasks: Task A identifies what a post is about, such as a water-utility topic, supporting incident monitoring, operational routing, trend analysis, and separation of urgent incidents from routine communication; Task B identifies what the utility is trying to do, such as warn, instruct, update, or reassure the public, supporting communication audits and tracking how incidents are communicated over time. The tasks are related but not interchangeable: a burst-pipe post may concern service disruption under Task A, while its Task B function may be a warning, an instruction to store water, a repair

update, or reassurance that service has been restored. Task A therefore captures the subject of the message, whereas Task B captures its communicative purpose, together supporting both operational monitoring and evaluation of public-risk communication.

Natural language processing (NLP) offers a way to analyze such communication at scale, but existing work does not yet fully address this setting (Bonenkamp et al., 2025; Kurfali et al., 2025). Crisis-NLP has developed datasets and methods for relevance, informativeness, actionability, and urgency in disaster social media, including multilingual settings (Kaufhold et al., 2020; Al Amer et al., 2023; Meunier et al., 2025). ClimateNLP has expanded benchmarks for climate-related classification, retrieval, and intercultural communication (Bonenkamp et al., 2025; Kurfali et al., 2025; Bird et al., 2024). However, institution-authored infrastructure communication differs from citizen-generated crisis streams: most messages are routine or educational, while warnings and operational incidents form a smaller but important subset. To our knowledge, existing multilingual benchmarks do not jointly model the main topic and communicative function of European water-utility social-media communication.

To address this gap, we introduce a multilingual dataset of Facebook posts from European water utilities and evaluate NLP models and large language models (LLMs) to classify both the main water-utility topic and the primary communicative function of each post. We ask:

- RQ1:** How well can NLP models and LLMs classify multilingual water-utility Facebook posts by main topic and primary communicative function?
- RQ2:** How does classification performance differ between native-language and English-translated inputs, and how does the effect of translation vary across languages?
- RQ3:** How do LLMs compare with classical baselines such as keyword rules, term frequency-inverse document frequency (TF-IDF), and multilingual embedding models?

Our label space covers water quality, contamination, service disruption, conservation, drought, flooding, wastewater, and related public communication, capturing both acute risk messaging and

longer-term climate-water-stress communication (Lyle et al., 2023; Nijhawan and Howard, 2022; Schewe et al., 2014). We evaluate models on native-language posts and English-translated inputs, motivated by evidence that translation-based multilingual evaluation may help some models but can also distort source-language performance (Al Amer et al., 2023; Enomoto et al., 2025; Hu et al., 2020).

Our contributions are: (1) a multilingual corpus of 20,000 institution-authored water-utility posts with a 1,500-post manually annotated benchmark; (2) complementary topic and communication-function classification tasks; and (3) a systematic comparison of rule-based, supervised, and prompted LLM approaches across native and translated inputs.

2 Related Studies

ClimateNLP has developed a growing set of domain-specific datasets and benchmarks for climate claims, adaptation and policy documents, sustainability disclosures, and scientific information extraction (Bonenkamp et al., 2025; Kurfali et al., 2025; Diggelmann et al., 2020; Morio and Manning, 2023; Pan et al., 2025; Stede and Patz, 2021; Planas et al., 2022; Ni et al., 2023; Schimanski et al., 2024). These resources demonstrate the value of domain-specific classification and information extraction, but they predominantly target reports, scientific documents, policy texts, or corporate claims. AquaCrisis instead focuses on short, institution-authored messages from infrastructure providers, where climate-sensitive topics such as drought and flooding coexist with operational and routine public communication.

Crisis informatics shows that social media can support situational awareness, emergency response, and public communication (Kaufhold et al., 2020; Alam et al., 2018; Kumar et al., 2025; Alam et al., 2021a,b). Related work has also applied text-mining methods to mobile emergency-communication applications, showing how user feedback can reveal communication needs and support early-warning and emergency-management systems (Shaik and Oussalah, 2024). CrisisBench, HumAID, CrisisMMD, cross-lingual crisis classification, and SPEED++ address informativeness, humanitarian categories, multimodal understanding, and multilingual event extraction (Al Amer et al., 2023; Alam et al., 2018, 2021a,b; Parekh et al., 2024). Instruction-tuned LLMs can also

perform competitively on crisis-post classification, although supervised baselines often remain stronger (McDaniel et al., 2024). Water-related social-media NLP is more limited: WaterMM introduced bilingual water-quality relevance classification, while later studies examined water-quality feedback, contamination discourse, public opinion, and surveillance (Andreadis et al., 2022; Ahmad et al., 2023; Ayub et al., 2022; Mix et al., 2020). Prior work has mainly focused on citizen-generated Twitter/X content or binary relevance filtering, leaving institution-authored Facebook communication and fine-grained communicative functions understudied (Sallaku et al., 2017; Martin, 2017; Cheng et al., 2022).

Our taxonomy also draws on research on drinking-water risk and crisis communication. Work on drinking-water notices distinguishes the underlying hazard or operational issue from the communicative action directed toward the public, such as warning, instructing, explaining, or reassuring (Day et al., 2024; Bradford et al., 2017; Rundblad et al., 2010). Social-mediated crisis-communication research likewise emphasizes that the form and purpose of institutional messages shape public response (Jin et al., 2014; Cheng et al., 2022). AquaCrisis does not directly implement a single existing theoretical taxonomy; rather, its labels synthesize these communication perspectives with water-utility practice and pilot inspection of the corpus. Task A captures the dominant subject of a post, while Task B captures its primary communicative action, allowing the same operational topic to be studied across warning, instruction, update, reassurance, education, and institutional communication.

Our work also builds on multilingual NLP and translation-based inference (Enomoto et al., 2025; Hu et al., 2020; Conneau et al., 2020; Barbieri et al., 2022; Liu et al., 2025). Multilingual encoders such as XLM-R and XLM-T provide strong representations for noisy cross-lingual short-text classification (Conneau et al., 2020; Barbieri et al., 2022). Translating inputs into English can benefit English-oriented models, but may lose linguistic or cultural nuance, and native-language specialization can outperform translation-based approaches (Enomoto et al., 2025; Liu et al., 2025; Mullappilly et al., 2023). These findings motivate our comparison of native-language and English-translated water-utility posts rather than assuming a single multilingual strategy is optimal.

3 Methodology

Figure 1 summarizes the AquaCrisis corpus-construction and evaluation workflow. We first collect and preprocess multilingual posts from public Facebook pages operated by water utilities. Each post is represented in two forms: its native-language text and an English translation produced with a zero-shot GPT-4.1-nano prompt. Details of the preprocessing and translation process are provided in Appendix A. A stratified sample of 1,500 posts is then selected for gold-label annotation under Task A (Main water-utility topic) and Task B (Primary communicative function). The resulting gold-standard dataset is used to evaluate both supervised baselines and LLMs. The experiments compare native and translated text representations and include rule-based methods, stratified five-fold cross-validation for supervised models, and zero- and five-shot prompting for LLMs.

3.1 Data collection and corpus construction

Table 1 summarizes the original Facebook collection. We construct a multilingual corpus of public Facebook posts from water utilities and water-service organizations in European cities, collected with Apify (Apify, 2026). The original collection contains 29,352 Facebook records from 12 public utility pages, spanning August 2009 to June 2026 and covering 11 language labels. After deduplicating by post ID and removing records without usable textual content, the final corpus contains 20,000 posts. For each post, we retain the original text, timestamp, source URL, media type, and engagement metadata, including reactions, comments, and shares. We focus on Facebook because utilities use it for longer-form public updates, operational notices, and community-facing communication. Details of the utility accounts are provided in Table 5 of Appendix D.

We model these posts as public communication about water quality, service disruption, drought and conservation, flooding and wastewater, and related utility functions, rather than as generic organizational text. This follows crisis-informatics work that labels social-media posts for operational categories such as informativeness, humanitarian need, and event structure (Alam et al., 2018, 2021a,b; Imran et al., 2016). It also places the dataset within ClimateNLP, where recent work emphasizes domain-specific corpora, climate benchmarks, and adaptation-oriented tasks (Kurfali et al., 2025; Liu

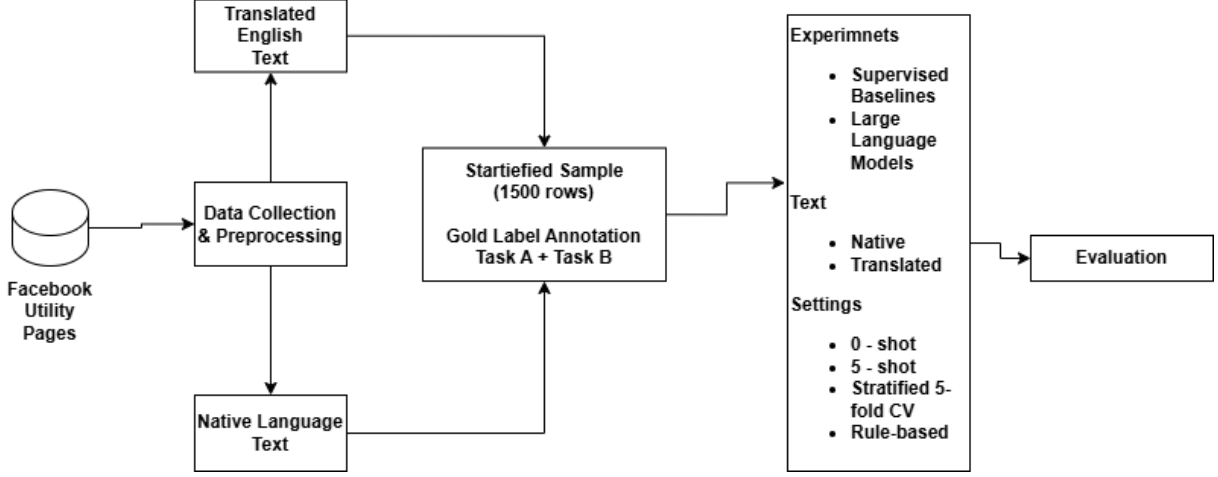


Figure 1: Overview of the AquaCrisis corpus-construction and evaluation workflow

Statistic	Original Facebook collection
Raw collected rows	29,352
Unique post IDs	20,883
Utility pages	12
Platform	Facebook
Date range	6 Aug. 2009–8 Jun. 2026
Explicit language labels	11 languages
Final text-bearing posts	20,000

Table 1: Statistics of the original collected Facebook dataset

et al., 2025; Webersinke et al., 2021). Our corpus extends this agenda to climate-sensitive infrastructure communication, where utilities communicate risk, reassurance, instruction, and service continuity to the public. The corpus is unevenly distributed across languages and utilities; Portuguese and Finnish together account for nearly half of the preprocessed posts, while several languages contribute substantially smaller subsets (Figure 3).

3.2 Annotation scheme and gold standard

We annotate a 1,500-post gold-standard subset in Label Studio (HumanSignal, 2026). Instead of uniform random sampling, we use a stratified, keyword-guided strategy to increase coverage of rare but operationally important posts. The keyword lists were derived from the same domain-informed taxonomy used for the classification tasks: we first defined categories from water-utility communication and climate-related water-risk knowledge, then selected keywords reflecting those cate-

gories. This follows crisis-dataset practice, where task-aware sampling helps capture informative but infrequent crisis content (Alam et al., 2021a). The full sampling lexicon and priority signals are reported in Table 8 in Appendix G. The resulting gold set is therefore an enriched benchmark sample rather than an estimate of category prevalence in the full corpus. Because sampling uses taxonomy-related lexical cues on English-translated text, performance, particularly for lexical methods such as translated TF-IDF, may also be affected by this enrichment.

Two domain-informed annotators reviewed the guidelines and completed a pilot round to clarify label definitions and difficult cases. They then independently assigned one Task A and one Task B label to each post. Before consensus, raw inter-annotator agreement was 72.3% for Task A and 79.1% for Task B, with Cohen’s κ of 0.672 and 0.715, respectively. Disagreements were resolved through discussion and consensus; unresolved cases were adjudicated by a third domain-informed annotator using the post, proposed labels, and guidelines. Annotators read the English translation and consulted the original-language text when the translation was unclear or potentially misleading. Detailed agreement statistics are reported in Table 4 of Appendix C.1, with the full annotation guidelines provided in Appendix C.

We define two main tasks. **Task A** asks, “What is the post mainly about?” and assigns the post’s primary water-utility topic: Water quality / safety / public health; Service disruption / repair / infrastructure; Water conservation / drought / demand management; Flood / stormwater / wastewater /

sewer; Environmental sustainability / waste / recycling / biodiversity; Public education / heritage / community engagement; and Routine / institutional / customer service / other. **Task B** asks, “What is the utility mainly trying to do through the post?” and assigns its primary communicative function: Alert / warning; Instruction / advice to public; Operational update / resolution; Reassurance / safety information; Education / awareness; Institutional promotion / community news; and Other / unclear.

Real posts may contain several topics or communicative functions. We nevertheless use one primary label per task to create a well-defined retrieval and routing benchmark: annotators select the topic that motivates the post or represents its main consequence, and the communicative function receiving the greatest emphasis or carrying the most important public-risk implication. This simplifies multi-function messages and should therefore be viewed as a primary-label representation rather than an exhaustive discourse annotation; multi-label annotation is a natural extension.

Because alerts, instructions, operational updates, and reassurance frequently overlap conceptually, we additionally derive **Task B.1** by mapping Alert / warning, Instruction / advice to public, Operational update / resolution, and Reassurance / safety information to Risk / incident communication while retaining education, institutional/community communication, and other/unclear as separate groups. B.1 is derived deterministically from Task B and provides a coarser evaluation of communicative role.

3.3 Baselines and multilingual classifiers

We evaluate the supervised baselines using stratified five-fold cross-validation on the annotated gold standard. In each fold, models are trained on approximately 80% of the labeled posts and evaluated on the held-out 20%, while preserving the label distribution where possible. We pool the out-of-fold predictions so that every annotated post is evaluated exactly once by a model that was not trained on that post. This provides a complete gold-set evaluation for comparison with the LLM results and reduces dependence on a single train-test split.

We include a majority-class baseline as a lower bound and a keyword-rule baseline as an interpretable monitoring approach. The keyword system assigns labels using manually curated lexical triggers associated with the task taxonomy and falls back to a default routine/other label when no rule

matches. The complete Task A and Task B lexicons are reported in Tables 9 and 10 in Appendix G.

Our second baseline is TF-IDF + logistic regression (LR). We compute unigram and bigram TF-IDF features and train a class-balanced logistic regression classifier on native and translated text variants. This setting is useful because short utility posts often contain recurring phrases such as water interruption, boil water, planned maintenance, heavy rain, or save water. Classical text classifiers remain strong references for short-text classification because they are fast, interpretable, and competitive on many domain-specific tasks (Joulin et al., 2017).

Our third baseline is a multilingual sentence-embedding classifier. We encode posts with paraphrase-multilingual-MiniLM-L12-v2 from Sentence-Transformers and train logistic regression on the dense embeddings. This follows the Sentence-BERT line of work, where semantically aligned sentence representations improve generalization across paraphrases and languages compared with purely lexical features (Reimers and Gurevych, 2019). Detailed baseline parameters are reported in Table 6 of Appendix E.

3.4 Zero-shot and few-shot LLM classification

We evaluate instruction-following LLMs with zero-shot and five-shot prompting on both original-language and English-translated posts. We use Gemma 4 (Gemma Team, 2026), GPT-4.1-mini (OpenAI, 2025), GPT-4o-mini (OpenAI, 2024), and Qwen 3.5 (Qwen Team, 2026) because they support multilingual inputs and enable direct prompted classification. The GPT-4o-mini setting is motivated by recent crisis-social-media work evaluating GPT-4o-family models for zero-shot crisis-tweet classification on CrisisBench, but we use the smaller model as a cost-efficient alternative for our larger native/translated and zero/five-shot benchmark design (McDaniel et al., 2024). For each post, the model predicts one Task A label and one Task B label. The prompt also requests a self-reported confidence value; this was collected as an exploratory diagnostic and was not used for prediction selection, filtering, or evaluation. We requested structured JSON output, but did not use grammar-constrained or guided decoding beyond the backend’s JSON-formatting mechanism.

This setup connects ClimateNLP and crisis-social-media NLP: the task is domain-specific, multilingual, and operational, but the adaptation

method is prompting rather than task-specific fine-tuning (Bonenkamp et al., 2025; Kurfali et al., 2025; Pan et al., 2025; Alam et al., 2021b). Zero-shot and few-shot prompting are standard adaptation settings for modern language models, but prior work shows that demonstrations, prompt order, and calibration can substantially affect stability (Zhao et al., 2021). We therefore treat five-shot prompting as a controlled experimental condition rather than assuming that adding examples will always improve performance. The full zero-shot and five-shot prompts are shown in Appendix B along with the prompt design details. Exact model identifiers, inference backends, and decoding parameters are reported in Table 7 of Appendix F.

3.5 Evaluation protocol

We report accuracy, macro-F1, weighted-F1, and per-class precision, recall, and F1 for both tasks. For the supervised baselines, the primary reported scores are computed from the pooled out-of-fold predictions across the five folds. Macro-F1 is central because the corpus contains frequent routine posts alongside rare but important categories such as contamination, service disruption, and warnings. Weighted-F1 complements this by reflecting performance under the observed label distribution. This metric bundle follows recent crisis-social-media benchmark practice (Al Amer et al., 2023; Alam et al., 2021a,b; Parekh et al., 2024). For supervised baselines, fold-level macro-F1 means and standard deviations are also provided as a complementary stability analysis.

Invalid generative predictions are retained rather than discarded. An invalid or out-of-schema Task A prediction is scored as incorrect for Task A, while an invalid Task B prediction is scored as incorrect for Task B and the derived Task B.1. When only one task prediction is invalid, the valid prediction for the other task is still evaluated normally. A malformed confidence value alone does not invalidate otherwise valid task labels. No gold examples are removed because of invalid model output. We additionally report the proportion of rows containing at least one invalid task label as a schema-compliance diagnostic.

To quantify uncertainty, we compute 95% percentile bootstrap confidence intervals using 10,000 stratified-bootstrap replicates. For each task, posts are resampled with replacement within the corresponding gold-label strata, and identical resampled post indices are used across systems for paired

comparisons. Macro-F1 is recomputed over the predefined label set on every replicate, with invalid predictions retained as incorrect. We describe evidence for a difference when the 95% paired interval for Δ macro-F1 excludes zero. Fold-level means and standard deviations for the supervised baselines are retained as a separate measure of cross-validation stability.

4 Results

4.1 Label distribution

Table 2 shows substantial class imbalance. Routine and institutional communication is the largest Task A category, while institutional news and education dominate Task B. Operational updates, reassurance, and unclear posts are especially rare, motivating our use of macro-F1 and the grouped Task B.1 evaluation in Table 3. The skew also helps explain why Task B is generally more difficult than Task A and why grouped Task B.1 improves performance.

4.2 Model performance

Table 3 reports macro-F1 scores on the gold standard. The majority baseline performs poorly (Task A: 0.069; Task B: 0.085), reflecting class imbalance. Keyword rules provide a stronger but limited baseline (0.562/0.309), while supervised models perform best on the fine-grained tasks. TF-IDF + LR on translated text achieves the highest Task A score (0.671), while multilingual embeddings + LR on translated text achieve the highest Task B score (0.504). Among LLMs, Gemma 4 reaches 0.658 on Task A, GPT-4.1-mini reaches 0.638, and GPT-4o-mini reaches 0.624. For Task B, GPT-4o-mini performs best among LLMs, reaching 0.392 with translated zero-shot prompting. Overall, task-specific supervised models remain strongest for both tasks A and B. Bootstrap intervals for individual systems are reported in Appendix H.4; paired comparisons are used below when interpreting differences between input conditions.

Task B is generally more difficult than Task A across the evaluated systems, plausibly because its labels are more ambiguous and imbalanced. Grouping sparse and overlapping Task B classes into Task B.1 improves most scores and particularly benefits GPT-4o-mini. Translated multilingual embeddings + LR rise from 0.504 to 0.592, while GPT-4o-mini reaches 0.660 with translated five-shot prompting and 0.650 with native five-shot prompting. This indicates that broader communicative-

Task	Category	Count	%
Task A	Water quality / safety / public health	65	4.3
Task A	Service disruption / repair / infrastructure	226	15.1
Task A	Water conservation / drought / demand management	140	9.3
Task A	Flood / stormwater / wastewater / sewer	136	9.1
Task A	Environmental sustainability / waste / recycling / biodiversity	209	13.9
Task A	Public education / heritage / community engagement	250	16.7
Task A	Routine / institutional / customer service / other	474	31.6
Task B	Alert / warning	205	13.7
Task B	Instruction / advice to public	164	10.9
Task B	Operational update / resolution	21	1.4
Task B	Reassurance / safety information	11	0.7
Task B	Education / awareness	446	29.7
Task B	Institutional promotion / community news	638	42.5
Task B	Other / unclear	15	1.0

Table 2: Label distribution in the 1,500-post gold-standard annotation sample for Task A and Task B.

Model	Text	Setting	Task A	Task B	Task B.1	Invalid Rows %
Majority	all	Stratified 5-fold CV	0.069	0.085	0.150	–
Keyword rules	translated	rule-based	0.562	0.309	0.356	–
TF-IDF + LR	translated	Stratified 5-fold CV	0.671	0.475	0.548	–
TF-IDF + LR	native	Stratified 5-fold CV	0.521	0.394	0.475	–
Multilingual embeddings + LR	translated	Stratified 5-fold CV	0.657	0.504	0.592	–
Multilingual embeddings + LR	native	Stratified 5-fold CV	0.649	0.495	0.575	–
Gemma 4	translated	zero-shot	0.658	0.352	0.563	0.00
Gemma 4	translated	five-shot	0.624	0.331	0.553	12.27
Gemma 4	native	zero-shot	0.633	0.346	0.552	0.00
Gemma 4	native	five-shot	0.566	0.292	0.497	19.80
GPT-4.1-mini	translated	zero-shot	0.638	0.359	0.494	0.00
GPT-4.1-mini	translated	five-shot	0.617	0.325	0.466	0.00
GPT-4.1-mini	native	zero-shot	0.616	0.328	0.491	0.07
GPT-4.1-mini	native	five-shot	0.596	0.315	0.463	0.00
GPT-4o-mini	translated	zero-shot	0.620	0.392	0.646	0.00
GPT-4o-mini	translated	five-shot	0.624	0.381	0.660	0.00
GPT-4o-mini	native	zero-shot	0.601	0.384	0.623	0.00
GPT-4o-mini	native	five-shot	0.605	0.372	0.650	0.00
Qwen 3.5	translated	zero-shot	0.567	0.277	0.472	0.13
Qwen 3.5	translated	five-shot	0.584	0.269	0.441	0.13
Qwen 3.5	native	zero-shot	0.558	0.277	0.461	0.53
Qwen 3.5	native	five-shot	0.573	0.242	0.430	1.20

Table 3: Model comparison on the 1,500-post gold standard. Scores are macro-F1. Supervised models are evaluated using pooled out-of-fold predictions from stratified five-fold cross-validation (CV). Keyword rules and LLMs are evaluated on the complete annotated sample. Invalid rows (%) denotes the percentage of LLM outputs with a missing, malformed, or out-of-schema Task A or Task B label. Bootstrap confidence intervals for the LLM systems are reported in Appendix H.4; paired baseline translation effects are reported in Section 4.3.

function categories are more robust than the original fine-grained labels and that LLMs become especially competitive when the label space is less granular. Additional accuracy, weighted-F1, class-wise results, fold-level variability, per-language comparisons, and bootstrap uncertainty analyses are reported in Tables 11–24 and Figures 4–9 of Appendix H.

LLMs also sometimes produced labels outside the predefined schema. The rightmost column of Table 3 reports invalid-output rates. Gemma 4 five-shot is the least stable, with 12.27% invalid outputs on translated text and 19.80% on native text. Qwen 3.5 remains at or below 1.20%, GPT-4.1-mini produces only one invalid native zero-shot row, and GPT-4o-mini produces no invalid rows. Five-shot prompting therefore does not consistently improve either performance or schema compliance. These

errors show that generative classification can return missing, malformed, or out-of-schema labels, so macro-F1 should be interpreted together with invalid-output rate, especially for operational use.

4.3 Translation versus native text

Figure 2 visualizes the native-versus-translated macro-F1 comparisons and their 95% bootstrap confidence intervals for Tasks A and B. Translation effects are strongly model-dependent. For TF-IDF + LR, translated input increases macro-F1 by 0.150 on Task A (95% paired confidence interval (CI) [0.124, 0.175]), by 0.080 on Task B ([0.037, 0.127]), and by 0.073 on Task B.1 ([0.055, 0.091]). In contrast, the translated–native differences for multilingual embeddings are small and their paired intervals include zero on all three tasks.

Among LLMs, translated input is higher for

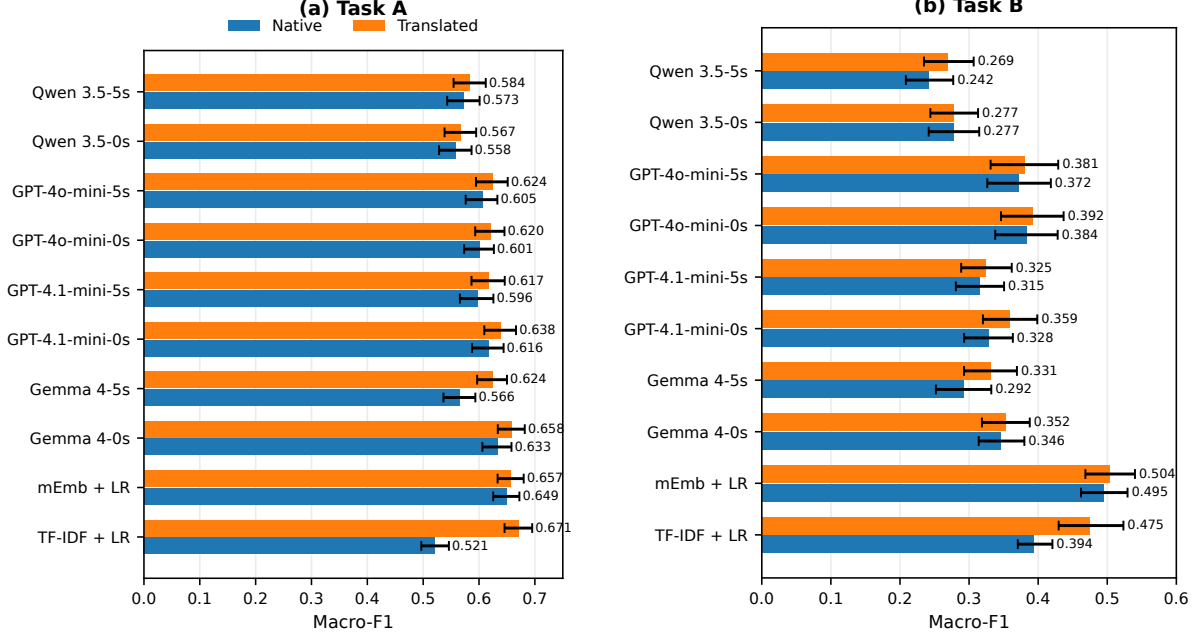


Figure 2: Native versus translated model performance on Tasks A and B. Bars show macro-F1 and whiskers show 95% percentile bootstrap confidence intervals from 10,000 stratified-bootstrap replicates. 0s and 5s denote zero-shot and five-shot prompting; mEmb = multilingual embeddings; LR = logistic regression. Paired LLM translated–native confidence intervals are reported in Appendix H.4

Gemma 4 on Task A in zero-shot prompting ($\Delta = 0.025$, [0.007, 0.045]) and on all three tasks under five-shot prompting ($\Delta = 0.058$, [0.033, 0.085] for Task A; 0.039, [0.004, 0.076] for Task B; and 0.056, [0.019, 0.092] for Task B.1). GPT-4.1-mini zero-shot also shows translated-input differences on Task A (0.022, [0.002, 0.043]) and Task B (0.032, [0.010, 0.054]). All translated–native intervals for GPT-4o-mini and Qwen 3.5 include zero. Translation therefore provides clear gains for the lexical classifier and selected LLM conditions, rather than a uniform advantage across models.

Per-language results are also heterogeneous (Appendix H.3). Among the eight languages with at least 50 gold posts, translated input gives numerically higher zero-shot Task A macro-F1 for Gemma 4 in all eight languages and for GPT-4.1-mini in seven of eight, whereas directions are less consistent for GPT-4o-mini and Qwen 3.5. Because several language subsets remain small and class coverage differs, we treat these results descriptively rather than as language-level significance tests.

5 Discussion

For RQ1, multilingual water-utility posts are classifiable, but performance depends strongly on the task. Task A yields higher scores than Task B for

all non-majority systems in our experiments, which may reflect the more concrete nature of the topic labels relative to the communicative-function distinctions. Translated TF-IDF has the highest Task A score, while translated multilingual embeddings perform best for Task B, showing that both lexical normalization and cross-lingual semantic representations are effective for short, domain-specific posts (Conneau et al., 2020; Barbieri et al., 2022; Stammbach et al., 2023; Schimanski et al., 2023). The improvement on grouped Task B.1 is consistent with broader communicative roles being easier to classify reliably than the fine-grained distinctions, particularly for rare labels such as reassurance and operational resolution (Mix et al., 2020; Andreadis et al., 2022; Stede and Patz, 2021; Schimanski et al., 2024).

For RQ2, translation effects are model- and setting-dependent rather than uniformly positive. English translation substantially benefits TF-IDF, consistent with lexical normalization across languages, whereas the multilingual embedding classifier shows small paired differences whose intervals include zero. Among LLMs, translated input shows evidence of gains for selected Gemma 4 and GPT-4.1-mini conditions, while the corresponding intervals for GPT-4o-mini and Qwen 3.5 in-

clude zero. The language-specific analysis further shows heterogeneous patterns across the eight sufficiently represented languages. These results support using translation as an experimental alternative rather than replacing native-language evaluation (Enomoto et al., 2025; Liu et al., 2025; Mullappilly et al., 2023).

For RQ3, LLMs are competitive but do not consistently outperform supervised baselines. Supervised models remain strongest on the fine-grained tasks, whereas GPT-4o-mini has the highest Task B.1 score, reaching 0.660 with translated five-shot prompting compared with 0.592 for the strongest supervised baseline. This suggests that instruction-tuned LLMs are particularly effective when the label space is broader, while task-specific classifiers retain an advantage for finer distinctions (McDaniel et al., 2024; Kurfali et al., 2025; Ni et al., 2023).

The LLM results also reveal practical limitations. Five-shot prompting does not provide consistent gains. For Gemma 4, five-shot Task A is lower than zero-shot on both native ($\Delta = -0.067$, 95% CI $[-0.094, -0.041]$) and translated input (-0.034 , $[-0.055, -0.013]$); several GPT-4.1-mini and Qwen comparisons are likewise lower, whereas all GPT-4o-mini five-shot–zero-shot intervals include zero. Nevertheless, generative classification remains sensitive to formatting and can return missing, malformed, or out-of-schema labels unless outputs are tightly constrained (McDaniel et al., 2024; Enomoto et al., 2025; Ni et al., 2023).

Water-utility communication remains challenging because posts are short, local, technical, multilingual, and often combine several communicative purposes. A single message may include infrastructure information, service status, advice, and reassurance, while small wording differences can carry important public-health consequences (Bradford et al., 2017; Rundblad et al., 2010). The main practical value of automated classification is therefore in triage and monitoring rather than full automation. Models can help identify alerts, contamination risks, drought, flooding, and service disruption, supporting situational awareness (Alam et al., 2018, 2021a,b; Planas et al., 2022).

Human oversight remains essential. Misclassifying a warning as routine, or reassurance as unrelated content, could affect public understanding, compliance, and trust (Day et al., 2024; Bradford et al., 2017; Rundblad et al., 2010). NLP should therefore be treated as decision support: models can filter and prioritize posts, but expert commu-

nicators should verify high-stakes cases. Overall, this study extends climate and crisis NLP to multilingual, institution-authored water-utility communication while keeping human judgment central (Andreadis et al., 2022; Ahmad et al., 2023; Ayub et al., 2022; Jin et al., 2014; Stede and Patz, 2021; Planas et al., 2022; Schimanski et al., 2024; Kumar et al., 2025).

6 Conclusion

We introduced a multilingual benchmark for classifying Facebook communication from European water utilities by topic and communicative function. The dataset captures both routine institutional messaging and climate-relevant communication about water quality, service disruption, drought, flooding, wastewater, sustainability, and public guidance. We evaluate fine-grained Task A and Task B labels together with the broader grouped Task B.1 categories.

The results show that water-utility topics are easier to classify than fine-grained communicative functions. Translated TF-IDF with logistic regression performs best on Task A, while translated multilingual sentence embeddings with logistic regression perform best on Task B. LLMs remain competitive but do not consistently surpass supervised baselines; GPT-4o-mini performs best on grouped Task B.1, reaching 0.660 with translated five-shot prompting. Translation strongly benefits the lexical TF-IDF classifier, whereas paired uncertainty estimates show more selective effects for LLMs: some Gemma 4 and GPT-4.1-mini conditions improve with translation, while GPT-4o-mini and Qwen 3.5 are comparatively stable. Generative classification also requires careful schema control, as Gemma 4 produces substantial invalid-output rates in its five-shot conditions.

Future work should move from post-level classification toward incident-centered analysis by linking related messages into event timelines and extracting affected locations, incident status, public advice, and resolution information. Connecting utility communication with water-quality, weather, sensor, and emergency records could support monitoring while retaining expert oversight for high-stakes interpretation.

Limitations

This study has several limitations. The corpus is limited to Facebook posts from European wa-

ter utilities. Facebook is useful for studying public-facing updates and community engagement, but it does not capture the full utility communication ecosystem, which may include websites, press releases, emergency-alert systems, X/Twitter, LinkedIn, SMS alerts, and local media. The results therefore reflect Facebook-based utility communication in European contexts and may not generalize to regions with different infrastructure, regulation, climate risks, or communication practices.

The corpus is not balanced across languages or utilities. Some languages and pages contribute many more posts than others, reflecting real activity patterns rather than a balanced multilingual benchmark design. This may affect supervised learning and evaluation, especially for lower-frequency languages, and means that native-versus-translated results should not be interpreted as equally strong evidence for every language.

Emergency and outbreak-related posts are rare in the corpus. Most posts concern routine updates, education, community engagement, or customer service, while contamination events, boil-water notices, urgent warnings, reassurance messages, and operational resolutions occur less often. These rare categories are highly important for crisis communication, but their low frequency limits the reliability of per-class estimates.

The translated condition depends on GPT-4.1-nano machine translation. Translation enables comparison between native-language and English-translated inputs, but it may introduce lexical, cultural, or pragmatic errors, especially for local infrastructure terms, place names, informal language, and risk-communication phrasing. Improvements on translated text should therefore not be read as evidence that translation is always preferable. Because annotators primarily reviewed the English translations and consulted the original-language text when needed, translation errors may also have influenced some gold labels, introducing potential annotation bias.

The manually annotated gold standard contains 1,500 posts. This is sufficient for an initial benchmark and model comparison, but it limits reliability for rare labels, and macro-F1 remains sensitive to minority-class performance. Future work should expand the annotation set, include more incident-focused examples, and consider hierarchical or multi-label schemes for posts combining risk topics, instructions, reassurance, and institutional communication.

Language-specific estimates are uneven in reliability because the gold set is dominated by a subset of languages and several languages contain few examples of rare Task B classes.

Finally, generative models were evaluated using JSON-formatting constraints but not stronger grammar-constrained or guided decoding; such methods may reduce schema violations and should be evaluated in future work.

Ethical Considerations

The corpus is restricted to public organizational posts made by institutional utility accounts, not private user data. We do not profile individual citizens, infer sensitive traits, or release comment-level personal data. Nevertheless, the corpus may contain place names, infrastructure details, and references to local incidents. The distribution of languages, utilities, and communication categories is uneven, and the dataset should not be treated as representative of all water utilities, regions, or communication platforms.

Data Availability The gold annotations, predefined cross-validation fold assignments, experimental code, prompts, model predictions, and derived evaluation results are publicly available at <https://github.com/vishal310197/Aquacrisis>.

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A Translation and text representations

Figure 3 shows the language distribution after pre-processing. Each post is represented in two forms: the original-language text and an English translation produced with a zero-shot GPT-4.1-nano prompt. This allows us to compare native-language inference with a translate-to-English pipeline. Because recent multilingual LLM work shows that English translation can help some models but is not uniformly optimal, especially when pragmatic or cultural cues matter, we treat translation as an experimental condition rather than a default preprocessing step (Liu et al., 2025). The full translation prompt can be found in Appendix B.1.

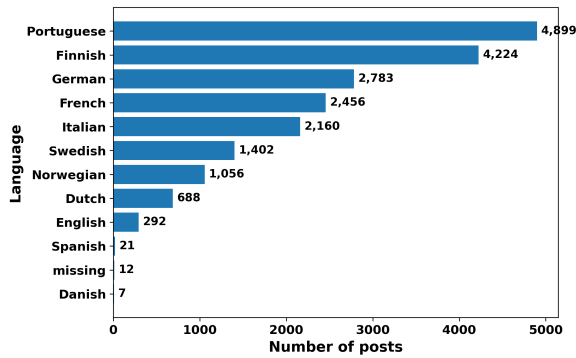


Figure 3: Language distribution in the preprocessed Facebook dataset.

For supervised baselines, we use normalized text to reduce lexical sparsity. The preprocessing pipeline unescapes HTML, normalizes Unicode, removes URLs and mentions, and lowercases text. Hashtag content is preserved as lexical material by converting forms such as #savewater into savewater. For LLM classification, we use the original post strings rather than cleaned text. This preserves punctuation, emojis, formatting, and discourse markers that may signal communicative function. Thus, classical baselines receive normalized task-oriented text, while LLMs receive the closest available version of the public-facing post.

B LLM Prompts

B.1 Translation Prompt

We translated non-English Facebook posts into English using GPT-4.1-nano through a batched OpenAI translation pipeline with temperature set to 0. The user message contained a JSON list of records with key, language, and text fields.

Listing 1: System prompt used for translating Facebook posts into English.

```
You are a professional translation
engine for public-sector social
media posts.
Translate each text into clear English.

Rules:
- Preserve numbers, dates, street names,
  place names, hashtags, URLs, and
  emojis.
- Do not summarize.
- Do not add information.
- If the text is already English, return
  it unchanged.
- If text is empty, return an empty
  string.
- Return ONLY valid JSON as a list of
  objects:
[{"key": "...", "translated_text":
  "..."}]
```

B.2 Zero-shot Classification Prompt

We used the following zero-shot prompt template for LLM classification. The model was asked to assign one label for Task A and one label for Task B, using only the predefined label set.

Listing 2: Zero-shot LLM classification prompt.

```
[SYSTEM MESSAGE]

You are a careful text classification
assistant. Your task is to assign
exactly one label for Task A and
exactly one label for Task B.

Follow these rules:
- Use only the label IDs and names
  listed below.
- Choose the single best label for each
  task.
- Do not output explanations.
- Return valid JSON only.
- If the post is short or ambiguous,
  choose the most plausible label
  using the definitions below.

Task A labels:

A1 Water quality / safety / public
  health
Posts about water safety, drinking-water
  quality, contamination, testing,
  boil-water notices, bacteria,
  chemical risks, turbidity, chlorine,
  PFAS, or health risk.

A2 Service disruption / repair /
  infrastructure
Posts about outages, interruptions,
  repairs, leaks, bursts, maintenance,
  restoration, crews, pipelines,
  treatment plants, or infrastructure
  problems.
```

A3 Water conservation / drought / demand management
Posts about saving water, drought, restrictions, efficient use, demand reduction, conservation campaigns, or scarcity.

A4 Flood / stormwater / wastewater / sewer
Posts about flooding, heavy rain response, drains, sewer overflow, stormwater, wastewater incidents, or sewage.

A5 Environmental sustainability / waste / recycling / biodiversity
Posts about environmental protection, emissions, habitat, recycling, biodiversity, wetlands, river ecology, sustainability programs, or nature stewardship.

A6 Public education / heritage / community engagement
Posts about educational activities, awareness events, tours, school visits, museum/heritage content, public exhibitions, or community engagement not mainly about a current operational issue.

A7 Routine / institutional / customer service / other
Posts about office hours, billing, job vacancies, holiday notices, generic greetings, routine institutional communication, or items that do not fit A1-A6.

Task B labels:

B1 Alert / warning
Urgent alert or warning about a risk, disruption, incident, hazard, or emergency.

B2 Instruction / advice to public
Tells people what to do, for example boil water, avoid an area, conserve water, prepare, report leaks, or follow safety advice.

B3 Operational update / resolution
Status update about work underway, service restoration, incident handling, repair progress, or completion.

B4 Reassurance / safety information
Explains that water is safe, risks are low, monitoring is ongoing, or provides calming safety clarification.

B5 Education / awareness
Explains information for public understanding without mainly promoting the institution.

B6 Institutional promotion / community

news
Promotes events, campaigns, achievements, awards, partnerships, or general community news.

B7 Other / unclear
Use when the communication function is too vague, ambiguous, context-dependent, or does not fit B1-B6.

Return this JSON schema exactly:

```
{
  "task_a": {
    "label_id": "...",
    "label_name": "..."
  },
  "task_b": {
    "label_id": "...",
    "label_name": "..."
  },
  "confidence": 0.00
}
```

[USER MESSAGE]

Classify this post:
{text}

B.3 Five-shot Classification Prompt

The five-shot condition used the same system instructions, label definitions, output schema, and classification rules as the zero-shot prompt in Listing 2. Five fixed demonstrations were then inserted before the final classification query, as shown below.

Listing 3: Five-shot LLM classification prompt with the five fixed demonstrations used in the experiments.

[USER MESSAGE]

Example 1
Text: Due to possible bacterial contamination, residents in the affected area must boil tap water before drinking or cooking until further notice.

Output:

[ASSISTANT MESSAGE]

```
{"task_a":{"label_id":"A1","label_name":"Water quality / safety / public health"},"task_b":{"label_id":"B2","label_name":"Instruction / advice to public"},"confidence":0.98}
```

[USER MESSAGE]

Example 2
Text: Repair crews are working on a burst water pipe. Water pressure is

low, and service is expected to return this evening.
Output:

[ASSISTANT MESSAGE]

```
{
  "task_a": {
    "label_id": "A2",
    "label_name": "Service disruption / repair / infrastructure"
  },
  "task_b": {
    "label_id": "B3",
    "label_name": "Operational update / resolution"
  },
  "confidence": 0.96
}
```

[USER MESSAGE]

Example 3
Text: During the dry and hot period, please save water by avoiding garden watering during the day and using water wisely.
Output:

[ASSISTANT MESSAGE]

```
{
  "task_a": {
    "label_id": "A3",
    "label_name": "Water conservation / drought / demand management"
  },
  "task_b": {
    "label_id": "B2",
    "label_name": "Instruction / advice to public"
  },
  "confidence": 0.95
}
```

[USER MESSAGE]

Example 4
Text: Heavy rain may overload drains and cause sewer overflow in low-lying areas. Avoid flooded streets and follow local warnings.
Output:

[ASSISTANT MESSAGE]

```
{
  "task_a": {
    "label_id": "A4",
    "label_name": "Flood / stormwater / wastewater / sewer"
  },
  "task_b": {
    "label_id": "B1",
    "label_name": "Alert / warning"
  },
  "confidence": 0.95
}
```

[USER MESSAGE]

Example 5
Text: Learn why correct waste sorting and recycling protect rivers, wildlife, and local biodiversity.
Output:

[ASSISTANT MESSAGE]

```
{
  "task_a": {
    "label_id": "A5",
    "label_name": "Environmental sustainability / waste / recycling / biodiversity"
  },
  "task_b": {
    "label_id": "B5",
    "label_name": "Education / awareness"
  },
  "confidence": 0.94
}
```

```
confidence": 0.94}
```

[USER MESSAGE]

Now classify this new post:
{text}

B.4 Prompt Design Notes

The prompts were designed to make translation and classification outputs constrained, reproducible, and easy to parse. For translation, the system prompt framed the model as a public-sector social-media translation engine and instructed it to produce clear English while preserving numbers, dates, street names, place names, hashtags, URLs, and emojis; it also prohibited summarization and added information, and required a valid JSON list as output. For classification, each prompt defined the task context as Facebook communication by European water utilities, listed the complete Task A and Task B label sets, and required the model to choose exactly one label for each task. We kept the prompt structure fixed across models, languages, and shot settings so that differences reflected the model, text source, and number of examples rather than changes in prompt wording.

C Annotation

C.1 Inter-annotator agreement

Agreement was calculated on the independent annotations before discussion or adjudication. The most frequent disagreements involved adjacent operational/topic categories for Task A and distinctions among education, institutional communication, instruction, alert, and operational update for Task B.

Task	<i>N</i>	Raw agreement	Cohen's κ
Task A	1500	72.3%	0.672
Task B	1500	79.1%	0.715

Table 4: Inter-annotator agreement before consensus resolution.

C.2 Guidelines

Annotators assigned exactly one Task A label and one Task B label to each post. Task A represents the main topic or water-related issue discussed in the post, whereas Task B represents the primary communicative function of the post. The two tasks were labeled independently: the Task A decision

did not automatically determine the Task B decision.

Task B.1 was not annotated separately. It was derived after annotation by grouping the fine-grained Task B labels into four broader communicative-function categories.

C.3 General Annotation Procedure

Annotators followed the decision process below.

1. Read the complete translated post, including its title, sentence context, hashtags, and any visible formatting.
2. Consult the original-language text when the translation was unclear, incomplete, or appeared to mistranslate a place name, infrastructure term, warning, or instruction.
3. Identify the main subject of the post and assign one Task A label.
4. Identify what the utility is mainly trying to accomplish communicatively and assign one Task B label.
5. When several labels were possible, select the label that was most central to the complete post and most important for public water-utility communication.
6. Use the routine or unclear labels only when none of the more specific labels adequately described the post.

Labels were based on the overall meaning of the post rather than the presence of an isolated keyword. For example, the word *water* alone did not justify a water-quality label, and the word *warning* did not automatically indicate an alert if the post was describing a historical event or educational campaign.

C.4 Distinguishing Task A and Task B

Task A answers:

What is the post mainly about?

Task B answers:

What is the utility mainly trying to do through this post?

A post may therefore receive different but complementary labels across the two tasks. For example, a post about a pipe burst may be labeled as service disruption for Task A, while its Task B label may be:

- *Alert / warning*, when it announces the disruption;
- *Instruction / advice to public*, when it tells residents to store water;
- *Operational update / resolution*, when it reports repair progress; or
- *Reassurance / safety information*, when it confirms that service has been restored or the water is safe.

Annotators therefore considered the topic and communicative function separately rather than assigning Task B solely from the Task A label.

C.5 Task A: Main water-utility topic

A1: Water quality / safety / public health

Use A1 when the main subject is the safety or quality of drinking water, contamination, testing, monitoring, or a possible health risk.

Typical indicators include:

- boil-water or do-not-drink notices;
- bacterial, chemical, or biological contamination;
- PFAS, chlorine, turbidity, colour, taste, or odour;
- drinking-water sampling and laboratory results;
- statements about whether tap water is safe;
- health effects related to water consumption.

Examples:

“Boil tap water before drinking because bacteria may be present.”

“Recent samples confirm that the water is safe to drink.”

Do not use A1 when water quality is only mentioned incidentally and the main issue is an outage, pipe repair, flooding, wastewater operation, or general environmental education.

A2: Service disruption / repair / infrastructure

Use A2 when the main subject is water-service continuity, network operation, repair work, maintenance, construction, or infrastructure failure.

Typical indicators include:

- water outages or interruptions;
- low pressure or temporary loss of supply;
- pipe leaks, bursts, or network faults;
- planned maintenance or construction;
- repair crews and restoration work;
- pipelines, pumping stations, treatment facilities, or other infrastructure.

Examples:

“Water supply will be interrupted tomorrow because of pipe maintenance.”

“Repair teams are working on a burst main causing low pressure.”

Use A2 even when the post also includes an instruction such as storing water. The instruction affects Task B, while the underlying topic remains service disruption.

A3: Water conservation / drought / demand management

Use A3 when the main subject is water scarcity, drought, high demand, restrictions, or reducing water consumption.

Typical indicators include:

- drought or prolonged dry weather;
- water-use restrictions;
- high summer demand or low reservoir levels;
- requests to reduce consumption;
- efficient household or garden water use;
- conservation and demand-management campaigns.

Examples:

“Please avoid watering gardens during the current dry period.”

“Small changes at home can reduce daily water demand.”

A concrete request to save water is normally labeled B2 for Task B. A post that mainly explains why conservation matters without directing the public to act is normally labeled B5.

A4: Flood / stormwater / wastewater / sewer

Use A4 when the main subject is flooding, rainfall response, stormwater, drains, wastewater, sewers, sewage, or wastewater treatment.

Typical indicators include:

- heavy rain and local flooding;
- stormwater drainage;
- sewer overflow or blockage;
- sewage discharge or wastewater incidents;
- wastewater-treatment processes or facilities;
- runoff, drainage maintenance, or combined sewer systems.

Examples:

“Heavy rain may overload drains and cause sewer overflow.”

“Wastewater-treatment operations have been affected by the storm.”

Use A4 rather than A2 when the post primarily concerns sewer, wastewater, drainage, or flood management, even when infrastructure repair is also mentioned.

A5: Environmental sustainability / waste / recycling / biodiversity

Use A5 when the main subject is environmental protection, sustainability, waste management, recycling, biodiversity, ecosystems, or nature stewardship.

Typical indicators include:

- recycling and waste sorting;
- hazardous, plastic, or food waste;
- circular-economy initiatives;
- biodiversity, wetlands, rivers, or habitats;
- emissions, energy efficiency, or climate initiatives;
- environmental restoration and protection.

Examples:

“Correct waste sorting helps prevent plastic from entering rivers.”

“The new wetland project supports local biodiversity.”

Use A5 when the environmental issue is the subject of the message. Use A6 when the post is mainly about a school visit, exhibition, tour, or community activity, even if the event includes environmental content.

A6: Public education / heritage / community engagement

Use A6 when the main subject is an educational, cultural, historical, or community-oriented activity related to water.

Typical indicators include:

- school visits and workshops;
- museum or treatment-plant tours;
- exhibitions and public-information events;
- water heritage and historical infrastructure;
- World Water Day activities;
- community participation not centred on a current incident.

Examples:

“Students visited the water museum to learn about the water cycle.”

“Join a guided tour of the city’s historic water tower.”

Use A6 when the activity itself is the central topic. Use A5 when the main subject is an environmental programme, ecosystem, sustainability policy, or waste-management issue.

A7: Routine / institutional / customer service / other

Use A7 for routine organizational communication or posts whose topic does not fit A1-A6.

Typical indicators include:

- billing, payment, or customer-service information;
- office hours and contact details;

- recruitment and job vacancies;
- holiday greetings or closure notices;
- awards, internal achievements, or staff news;
- general institutional announcements;
- content unrelated to a specific water-risk category.

Examples:

“Our customer-service office will be closed during the holiday.”

“We are recruiting a new project manager.”

A7 is the fallback Task A category, but it should not be used when a specific operational, public-health, drought, wastewater, environmental, or educational topic is clearly present.

C.6 Task B: Primary communicative function

B1: Alert / warning

Use B1 when the primary function is to notify the public of an urgent or potentially harmful situation.

Typical indicators include:

- warnings about contamination, outages, flooding, or hazards;
- announcements of an unexpected incident;
- urgent notices using language such as *warning*, *attention*, *emergency*, or *until further notice*;
- information whose main purpose is to make the public aware that a risk or disruption exists.

Examples:

“Warning: water supply has been interrupted in the northern district.”

“Heavy rain may cause flooding in low-lying areas.”

When a post both announces an incident and gives a clear instruction, use B2 if the requested action is the dominant communicative purpose. Use B1 when notifying the public of the risk is primary and any advice is secondary.

B2: Instruction / advice to public

Use B2 when the main function is to tell people what action to take.

Typical indicators include:

- boil, avoid, store, conserve, report, prepare, or evacuate;
- safety precautions;
- requests to reduce water use;
- instructions for reporting leaks;
- waste-sorting or disposal directions;
- steps to follow before, during, or after an incident.

Examples:

“Boil water for at least one minute before drinking.”

“Please store enough water before tomorrow’s interruption.”

General educational tips may also contain imperative wording. Use B2 when the post asks the public to perform a specific action. Use B5 when the main purpose is explanation or awareness rather than immediate behaviour.

B3: Operational update / resolution

Use B3 when the primary function is to report the status, progress, or resolution of operational work or an incident.

Typical indicators include:

- repair work is underway;
- crews are on site;
- expected restoration times;
- changes in service status;
- confirmation that a fault has been repaired;
- confirmation that service has returned.

Examples:

“Repairs are continuing, and service is expected to return by 18:00.”

“The pipe fault has been repaired and water service is restored.”

Use B3 when the post primarily reports what the utility is doing or what stage the incident has reached. An initial announcement of a new problem is normally B1, while a request for public action is normally B2.

B4: Reassurance / safety information

Use B4 when the primary function is to reduce concern or clarify that a risk is controlled, limited, or absent.

Typical indicators include:

- confirmation that water is safe;
- negative contamination results;
- statements that monitoring is continuing;
- explanations that unusual taste, colour, or smell is harmless;
- statements intended to calm concern or maintain confidence.

Examples:

“Tests found no contamination, and the water is safe to drink.”

“The unusual taste is temporary and does not present a health risk.”

Use B4 when safety clarification or reassurance is the main purpose. Use B3 when the main emphasis is repair progress or restoration of service.

B5: Education / awareness

Use B5 when the post mainly explains a topic, provides general knowledge, or raises awareness.

Typical indicators include:

- explanations of water treatment or wastewater processes;
- facts about conservation, recycling, climate, or water quality;
- educational campaigns;
- general advice presented primarily as awareness material;
- informational content without a current incident or direct operational request.

Examples:

“Learn how wastewater is treated before being returned to the environment.”

“This campaign explains why recycling protects waterways.”

Use B5 when teaching or awareness is the main purpose. Use B6 when the main purpose is promoting an event, partnership, achievement, or the utility itself.

B6: Institutional promotion / community news

Use B6 when the main function is to promote the utility, an event, campaign, partnership, achievement, or general community activity.

Typical indicators include:

- event invitations;
- awards and achievements;
- partnerships and sponsorships;
- recruitment or staff announcements;
- celebrations and holiday greetings;
- community news and organizational promotion.

Examples:

“Visit our information stand at the city festival this weekend.”

“We are proud to announce a new partnership with the local university.”

A campaign may contain educational material. Use B6 when promoting the campaign, organization, or event is dominant; use B5 when explaining the substantive issue is dominant.

B7: Other / unclear

Use B7 only when the communicative function cannot be assigned confidently.

Appropriate cases include:

- extremely short or context-free messages;
- unclear or incomplete translations;
- posts consisting mainly of a slogan, image reference, or external link;
- posts with no identifiable warning, instruction, update, reassurance, educational, or promotional function.

Examples:

“Water is life.”

“More information coming soon.”

Annotators were instructed not to use B7 merely because a post contained several functions. In such cases, they selected the dominant function using the rules below.

C.7 Rules for Ambiguous and Multi-Function Posts

Topic plus instruction. When a post describes an issue and tells the public what to do, Task A captures the issue and Task B normally captures the instruction.

“A pipe burst has interrupted supply.
Please store water for the evening.”

Task A: A2; Task B: B2.

Incident announcement versus operational update. Use B1 when the post mainly announces a new or ongoing risk. Use B3 when it mainly reports repair progress, expected restoration, or resolution.

Instruction versus education. Use B2 when the public is asked to perform a clear action. Use B5 when the post mainly explains why an action or issue matters.

Operational update versus reassurance. Use B3 when the focus is the status of work or service. Use B4 when the focus is safety, risk reduction, or calming public concern.

Education versus institutional promotion. Use B5 when the substantive information is central. Use B6 when the main purpose is to advertise an event, campaign, partnership, or organizational achievement.

Current event versus historical or educational content. Words such as *flood*, *drought*, or *contamination* were interpreted in context. A museum post about a historical flood, for example, could be labeled A6 rather than A4 when the main purpose was heritage or public education.

Multiple operational topics. When several Task A categories were present, annotators selected the topic that motivated the post or created the most important water-related consequence. For example, a pipe failure causing temporary discoloured water

could be A2 if the post mainly concerned repair and service interruption, but A1 if the central message was a drinking-water safety warning.

Hashtags and links. Hashtags were treated as supporting context rather than decisive evidence. External links were not assumed to determine the label unless the surrounding post text clearly described their purpose.

Dominant-function principle. When a post contained several communicative functions, annotators selected the function receiving the greatest emphasis, appearing in the main statement, or carrying the most important public-risk implication. The presence of several functions alone did not justify B7.

C.8 Derived Grouped Task B.1

Task B.1 was generated deterministically from the final Task B labels:

- B1, B2, B3, and B4 were grouped as *Risk / incident communication*;
- B5 remained *Education / awareness*;
- B6 became *Institutional / community communication*;
- B7 remained *Other / unclear*.

This grouping was introduced because real utility posts frequently combine warnings, instructions, operational updates, and reassurance. Task B retains these distinctions for fine-grained evaluation, while Task B.1 evaluates the broader communicative role.

D Utility Accounts

Table 5 summarizes the public Facebook utility accounts included in the corpus. For each account, we report the assigned utility identifier, inferred utility or account name, associated city or region, number of posts, dominant language, and temporal coverage. The table shows that the corpus covers multiple European cities and languages, but is unevenly distributed across utilities, reflecting differences in Facebook activity and available timeline length.

E Baseline Experiment Settings

Table 6 summarizes the main baseline settings used in the supervised and rule-based experiments.

These settings document the text variants, model parameters, and evaluation protocol used for the non-LLM baselines.

F LLM configurations

Table 7 summarizes the model identifiers, inference backends, decoding settings, and evaluation conditions used for the LLM experiments. For the local Ollama runs, the exact runtime build and quantization metadata were not preserved. We therefore report the model identifiers used at inference rather than inferring unrecorded implementation details.

G Sampling and Keyword-Rule Lexicons

This appendix reports the complete English lexicons used for two distinct purposes: keyword-guided sampling of the annotation set and the deterministic keyword-rule classification baseline. The sampling lexicon was used only to increase coverage of domain-relevant posts and did not determine gold labels. Tables 8–10 report the sampling and baseline lexicons.

Because the sampling lexicon and keyword baseline were derived from the same domain taxonomy and share lexical cues, the keyword baseline should be interpreted as an in-domain lexical reference rather than a fully independent baseline.

Utility ID	Inferred utility or account	City/region	Posts	Dominant language	Coverage
U03	EPAL Águas de Lisboa	Lisbon	4,973	Portuguese	2011-2026
U01	HSY	Helsinki region	3,017	Finnish	2014-2026
U08	Eau de Paris	Paris	2,476	French	2009-2026
U09	ABC Napoli	Naples	2,164	Italian	2017-2026
U05	Berliner Wasserbetriebe	Berlin	1,550	German	2014-2026
U10	Stockholm Vatten och Avfall	Stockholm	1,398	Swedish	2009-2026
U02	Hamburg Wasser	Hamburg	1,243	German	2018-2026
U04	Oslo Vann	Oslo	1,062	Norwegian	2013-2026
U12	Turun Vesihuolto	Turku	744	Finnish	2016-2026
U07	Farys	Ghent	681	Dutch	2014-2026
U11	Tampereen Vesi	Tampere	466	Finnish	2020-2026
U06	World Waternet	Amsterdam	226	English	2023-2026

Table 5: Utility accounts included in the Facebook corpus. Post counts refer to text-bearing posts used in the dataset after preprocessing. Coverage indicates the year range represented for each account.

Baseline	Setting	Value
All trainable baselines	Evaluation protocol	Stratified five-fold cross-validation on the 1,500-post gold-standard sample
All trainable baselines	Fold construction	StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
All trainable baselines	Score aggregation	Predictions from the five held-out folds are pooled so that each gold-standard post is evaluated exactly once by a model that was not trained on that post
All trainable baselines	Primary reported scores	Metrics computed from the pooled out-of-fold predictions; fold-level means and standard deviations are also retained for assessing variability
All trainable baselines	Random seed	42
Majority baseline	Prediction rule	Predicts the most frequent label in the training portion of each fold
Majority baseline	Evaluation	Applied independently within each of the five cross-validation folds and evaluated using pooled out-of-fold predictions
Keyword rules	Text input	translated_clean
Keyword rules	Matching rule	Assigns the first label whose manually curated lexical triggers match the post text
Keyword rules	Default label, Task A	<i>Routine / institutional / customer service / other</i>
Keyword rules	Default label, Task B	<i>Other / unclear</i>
Keyword rules	Evaluation	Fixed rules require no training and are evaluated on the held-out portion of each fold; predictions are pooled across all five folds
TF-IDF + LR	Text variants	native_clean and translated_clean
TF-IDF + LR	Features	Word unigram and bigram TF-IDF features, ngram_range=(1, 2)
TF-IDF + LR	Maximum features	30,000
TF-IDF + LR	Minimum document frequency	2
TF-IDF + LR	Classifier	Logistic regression
TF-IDF + LR	Logistic-regression parameters	max_iter=2000, class_weight=balanced, and random_state=42
TF-IDF + LR	Fold fitting	The TF-IDF vectorizer and logistic-regression classifier are fitted only on the training portion of each fold
Multilingual embeddings + LR	Text variants	native_clean and translated_clean
Multilingual embeddings + LR	Encoder	sentence-transformers/ paraphrase-multilingual-MiniLM-L12-v2
Multilingual embeddings + LR	Embedding settings	batch_size=64 and normalize_embeddings=True
Multilingual embeddings + LR	Classifier	Logistic regression with max_iter=2000, class_weight=balanced, and random_state=42
Multilingual embeddings + LR	Fold fitting	The sentence encoder remains fixed; a new logistic-regression classifier is fitted on the training embeddings in each fold
Task B.1	Evaluation	Task B.1 predictions are obtained post hoc by mapping the Task B gold labels and out-of-fold predictions to the four grouped communication functions; no separate Task B.1 classifier is trained

Table 6: Baseline experiment settings for the majority, keyword-rule, TF-IDF logistic-regression, and multilingual sentence-embedding classifiers. Trainable baselines are evaluated using stratified five-fold cross-validation, and their primary scores are computed from pooled out-of-fold predictions.

Model	Identifier, size, and backend	Inference settings	Evaluation conditions
Gemma 4	gemma4:e4b; E4B effective-parameter variant; Ollama	Temperature = 0; JSON output; reasoning disabled (think=false); context window = 4096; max. generation = 256 tokens	Native and English-translated; zero-shot and five-shot
Qwen 3.5	qwen3.5:9b; 9B parameters; Ollama	Temperature = 0; JSON output; reasoning disabled (think=false); context window = 4096; max. generation = 256 tokens	Native and English-translated; zero-shot and five-shot
GPT-4.1-mini	gpt-4.1-mini; size not publicly disclosed; OpenAI API	Temperature = 0; JSON-object output	Native and English-translated; zero-shot and five-shot
GPT-4o-mini	gpt-4o-mini; size not publicly disclosed; OpenAI API	Temperature = 0; JSON-object output	Native and English-translated; zero-shot and five-shot

Table 7: LLM configurations used for classification. Each configuration predicts one Task A label and one Task B label for each of the 1,500 gold-standard posts. Task B.1 is derived post hoc by mapping Task B predictions to the grouped label space.

Sampling group	Keywords and phrases
Water quality / safety / contamination	water quality; drinking water; tap water; safe to drink; water safety; contamination; contaminated; pollution; turbidity; chlorine; bacteria; e coli; ecoli; coliform; legionella; pfas; boil water; boil; water sampling; water monitoring; quality monitoring; public health; health risk; beach quality; water test; water testing; water analysis
Service disruption / repair / infrastructure	water interruption; service interruption; water service interruption; interruption; disruption; outage; no water; water cut; low pressure; pressure; pipe burst; burst pipe; broken pipe; leak; fault; repair; maintenance; construction; water network; network; restoration time; water supply interruption; service alert; planned maintenance; emergency repair
Flood / stormwater / wastewater / sewer	flood; flooding; storm; stormwater; heavy rain; rainfall; rainwater; runoff; drainage; drain; sewer; sewerage; wastewater; wastewater treatment; overflow; combined sewer overflow; sewage; treatment plant
Water conservation / drought / demand management	drought; scarcity; water scarcity; water restriction; water restrictions; water conservation; water saving; save water; conserve water; water consumption; water demand; heatwave; heat wave; dry spell; sustainable water use; conservation tips; watering ban; water wisely
Environmental sustainability / waste / recycling / biodiversity	waste sorting; waste management; waste collection; recycling; hazardous waste; plastic waste; food waste; biowaste; bio waste; battery; circular economy; biodiversity; wildlife; ecosystem; baltic sea; air quality; environmental conservation; sustainability; climate change; invasive species; compost; organic waste
Education / awareness / campaign	education; educational; awareness; campaign; outreach; school; children; student; students; workshop; training; tour; museum; exhibition; world water day; tips; public awareness; learn; learning; webinar
Routine / institutional / community	holiday; christmas; greeting; celebration; anniversary; award; event; community engagement; community events; partnership; customer service; billing; bill; payment; job; recruitment; career; contest; cultural; heritage; promotion; brand; opening hours; office closed
Rare/crisis prioritization signal	boil; contaminat; bacteria; e coli; ecoli; legionella; pfas; outbreak; illness; turbidity; overflow; flood; drought; restriction; outage; interruption; pipe burst; burst pipe; broken pipe; low pressure; no water; sewer; wastewater

Table 8: English keyword groups used for constructing the keyword-guided annotation sample. Posts were matched on cleaned English-translated text using case-insensitive matching. If a post matched several groups, the first group in the predefined priority order was assigned. The final row lists terms used to prioritize rare or crisis-related candidates within each sampling group.

Task A label	Keyword rules
A1: Water quality / safety / public health	water quality; drinking water; tap water; safe to drink; water safety; contamination; contaminated; pollution; turbidity; chlorine; bacteria; e coli; ecoli; coliform; legionella; pfas; boil water; boil; water sampling; water monitoring; quality monitoring; public health; health risk; water test; water testing
A2: Service disruption / repair / infrastructure	water interruption; service interruption; interruption; disruption; outage; no water; water cut; low pressure; pipe burst; burst pipe; broken pipe; leak; fault; repair; maintenance; construction; water network; restoration time; planned maintenance; emergency repair
A3: Water conservation / drought / demand management	drought; scarcity; water scarcity; water restriction; water restrictions; water conservation; water saving; save water; conserve water; water consumption; water demand; heatwave; heat wave; dry spell; watering ban
A4: Flood / stormwater / wastewater / sewer	flood; flooding; stormwater; storm water; heavy rain; rainfall; runoff; drainage; sewer; sewerage; wastewater; wastewater treatment; overflow; sewage; treatment plant
A5: Environmental sustainability / waste / recycling / biodiversity	waste sorting; waste management; waste collection; recycling; hazardous waste; plastic waste; food waste; biowaste; battery; circular economy; biodiversity; wildlife; ecosystem; air quality; environmental conservation; sustainability; climate change; invasive species; compost
A6: Public education / heritage / community engagement	education; educational; awareness; campaign; outreach; school; children; student; workshop; training; tour; museum; exhibition; world water day; public awareness; heritage; history
A7: Routine / institutional / customer service / other	holiday; christmas; greeting; celebration; anniversary; award; event; partnership; customer service; billing; bill; payment; job; recruitment; career; contest; opening hours; office closed

Table 9: Task A keyword rules used by the translated-text keyword baseline. Rules were evaluated in label order from A1 to A7 using case-insensitive substring matching. The first matching label was returned; posts with no match defaulted to A7 (*Routine / institutional / customer service / other*).

Task B label	Keyword rules
B1: Alert / warning	warning; alert; urgent; emergency; do not drink; do not use; risk; danger; unsafe; interruption; outage; flood warning; service alert
B2: Instruction / advice to public	please; boil; boil water; save water; conserve water; avoid; do not; report; contact; prepare; store water; follow; use less; sort; recycle; bring; check
B3: Operational update / resolution	update; updated; ongoing; resolved; fixed; restored; completed; repair work; restoration; back to normal; service has returned; work continues
B4: Reassurance / safety information	safe; safe to drink; no risk; no contamination; monitoring; tested; test results; under control; normal; quality is good
B5: Education / awareness	learn; education; educational; awareness; did you know; tips; guide; explains; how; why; campaign; workshop; school; students
B6: Institutional promotion / community news	event; join us; celebration; holiday; award; proud; partnership; recruitment; job; career; museum; exhibition; festival; community
B7: Other / unclear	No explicit keywords. Used as the fallback when none of the B1-B6 rules matched.

Table 10: Task B keyword rules used by the translated-text keyword baseline. Rules were evaluated in label order from B1 to B6 using case-insensitive substring matching. The first matching label was returned; posts with no match defaulted to B7 (*Other / unclear*).

H Additional Evaluation Metrics

use identical resampled post indices for the native and translated conditions.

H.1 Overall accuracy and weighted-F1

The main paper reports macro-F1 as the primary metric because the gold-standard label distributions are imbalanced. For completeness, Tables 11 and 12 report overall accuracy and weighted-F1.

H.2 Five-fold stability of supervised baselines

Table 13 reports the mean and sample standard deviation of macro-F1 across the five held-out folds for each supervised baseline. These values complement the pooled out-of-fold scores reported in the main paper by showing fold-to-fold variability.

H.3 Per-language native-translated analysis

To examine RQ2 by language, we report results for languages represented by at least 50 posts in the gold set: Portuguese, Finnish, Italian, German, Swedish, Norwegian, French, and Dutch. English ($n = 5$) and Spanish ($n = 1$) are omitted from separate performance estimates because their samples are too small. Danish and records with missing language metadata had no examples in the gold set. Within each language, macro-F1 is computed over labels represented in that language subset. These analyses are descriptive because sample sizes and class coverage differ substantially across languages. Tables 14 and 15 report the corresponding translated–native differences for Tasks A and B.

H.4 Bootstrap confidence intervals

Tables 16 and 17 report uncertainty estimates from 10,000 stratified-bootstrap replicates with random seed 42. Individual intervals are 95% percentile bootstrap confidence intervals. Paired differences

Model	Text	Setting	Task A	Task B	Task B.1	Invalid Rows %
Majority	all	Stratified 5-fold CV	0.316	0.425	0.425	–
Keyword rules	translated	rule-based	0.562	0.387	0.422	–
TF-IDF + LR	translated	Stratified 5-fold CV	0.685	0.717	0.728	–
TF-IDF + LR	native	Stratified 5-fold CV	0.555	0.622	0.633	–
Multilingual embeddings + LR	translated	Stratified 5-fold CV	0.669	0.665	0.685	–
Multilingual embeddings + LR	native	Stratified 5-fold CV	0.662	0.643	0.667	–
Gemma 4	translated	zero-shot	0.646	0.552	0.689	0.00
Gemma 4	translated	five-shot	0.579	0.434	0.576	12.27
Gemma 4	native	zero-shot	0.626	0.535	0.674	0.00
Gemma 4	native	five-shot	0.532	0.399	0.527	19.80
GPT-4.1-mini	translated	zero-shot	0.653	0.501	0.612	0.00
GPT-4.1-mini	translated	five-shot	0.647	0.482	0.595	0.00
GPT-4.1-mini	native	zero-shot	0.635	0.481	0.604	0.07
GPT-4.1-mini	native	five-shot	0.631	0.470	0.589	0.00
GPT-4o-mini	translated	zero-shot	0.618	0.601	0.742	0.00
GPT-4o-mini	translated	five-shot	0.643	0.588	0.723	0.00
GPT-4o-mini	native	zero-shot	0.602	0.584	0.728	0.00
GPT-4o-mini	native	five-shot	0.623	0.575	0.714	0.00
Qwen 3.5	translated	zero-shot	0.606	0.458	0.603	0.13
Qwen 3.5	translated	five-shot	0.612	0.484	0.616	0.13
Qwen 3.5	native	zero-shot	0.587	0.448	0.589	0.53
Qwen 3.5	native	five-shot	0.592	0.454	0.600	1.20

Table 11: Overall accuracy on the 1,500-post gold standard. Supervised baseline scores are computed from pooled out-of-fold predictions under stratified five-fold cross-validation. Keyword rules and LLMs are evaluated on all 1,500 annotated posts. Invalid Rows % denotes the percentage of LLM outputs with a missing, malformed, or out-of-schema Task A or Task B label.

Model	Text	Setting	Task A	Task B	Task B.1	Invalid Rows %
Majority	all	Stratified 5-fold CV	0.152	0.254	0.254	–
Keyword rules	translated	rule-based	0.576	0.414	0.426	–
TF-IDF + LR	translated	Stratified 5-fold CV	0.690	0.717	0.727	–
TF-IDF + LR	native	Stratified 5-fold CV	0.558	0.626	0.630	–
Multilingual embeddings + LR	translated	Stratified 5-fold CV	0.673	0.682	0.686	–
Multilingual embeddings + LR	native	Stratified 5-fold CV	0.667	0.663	0.665	–
Gemma 4	translated	zero-shot	0.641	0.547	0.674	0.00
Gemma 4	translated	five-shot	0.609	0.442	0.588	12.27
Gemma 4	native	zero-shot	0.625	0.536	0.662	0.00
Gemma 4	native	five-shot	0.582	0.417	0.558	19.80
GPT-4.1-mini	translated	zero-shot	0.647	0.511	0.617	0.00
GPT-4.1-mini	translated	five-shot	0.645	0.463	0.576	0.00
GPT-4.1-mini	native	zero-shot	0.630	0.485	0.605	0.07
GPT-4.1-mini	native	five-shot	0.628	0.447	0.565	0.00
GPT-4o-mini	translated	zero-shot	0.608	0.602	0.741	0.00
GPT-4o-mini	translated	five-shot	0.639	0.556	0.710	0.00
GPT-4o-mini	native	zero-shot	0.587	0.583	0.725	0.00
GPT-4o-mini	native	five-shot	0.617	0.547	0.703	0.00
Qwen 3.5	translated	zero-shot	0.603	0.447	0.595	0.13
Qwen 3.5	translated	five-shot	0.609	0.431	0.567	0.13
Qwen 3.5	native	zero-shot	0.585	0.440	0.584	0.53
Qwen 3.5	native	five-shot	0.591	0.404	0.564	1.20

Table 12: Overall weighted-F1 on the 1,500-post gold standard. Supervised baseline scores are computed from pooled out-of-fold predictions under stratified five-fold cross-validation. Keyword rules and LLMs are evaluated on all 1,500 annotated posts. Invalid Rows % denotes the percentage of LLM outputs with a missing, malformed, or out-of-schema Task A or Task B label.

Model	Text	Task A	Task B	Task B.1
Majority	all	0.069 $\pm$ 0.000	0.085 $\pm$ 0.000	0.149 $\pm$ 0.000
Keyword rules	translated	0.563 $\pm$ 0.021	0.307 $\pm$ 0.021	0.356 $\pm$ 0.025
TF-IDF + LR	translated	0.672 $\pm$ 0.028	0.464 $\pm$ 0.054	0.548 $\pm$ 0.012
TF-IDF + LR	native	0.520 $\pm$ 0.039	0.395 $\pm$ 0.029	0.475 $\pm$ 0.024
Multilingual embeddings + LR	translated	0.656 $\pm$ 0.030	0.502 $\pm$ 0.053	0.587 $\pm$ 0.052
Multilingual embeddings + LR	native	0.650 $\pm$ 0.038	0.492 $\pm$ 0.046	0.571 $\pm$ 0.043

Table 13: Mean macro-F1 and sample standard deviation across the five held-out folds. Each fold contains 300 evaluation posts. These values describe fold-to-fold variability; the main model-comparison table reports macro-F1 computed from the pooled out-of-fold predictions over all 1,500 posts.

Language (n)	Gemma 4		GPT-4.1-mini		GPT-4o-mini		Qwen 3.5	
	0s	5s	0s	5s	0s	5s	0s	5s
Portuguese (420)	+0.009	+0.040	+0.012	+0.022	-0.006	+0.022	-0.028	+0.012
Finnish (363)	+0.008	+0.086	+0.003	-0.020	+0.018	+0.048	+0.022	+0.020
Italian (201)	+0.151	+0.029	+0.073	+0.196	+0.010	-0.067	+0.135	+0.063
German (191)	+0.062	0.000	+0.042	+0.073	+0.039	+0.006	+0.078	+0.049
Swedish (97)	+0.096	+0.102	+0.120	+0.134	-0.009	-0.016	+0.033	+0.056
Norwegian (83)	+0.054	+0.005	+0.013	+0.068	+0.004	-0.024	-0.011	-0.058
French (83)	+0.061	+0.005	-0.056	-0.054	+0.150	-0.095	-0.061	-0.016
Dutch (56)	+0.077	-0.098	+0.102	+0.008	+0.004	+0.045	+0.161	-0.037

Table 14: Per-language translated–native differences in Task A macro-F1. Values are $\Delta = F1_{\text{translated}} - F1_{\text{native}}$, so positive values indicate higher performance on English-translated input. 0s and 5s denote zero-shot and five-shot prompting. Results are shown for languages with at least 50 posts in the gold set.

Language (n)	Gemma 4		GPT-4.1-mini		GPT-4o-mini		Qwen 3.5	
	0s	5s	0s	5s	0s	5s	0s	5s
Portuguese (420)	-0.010	+0.022	-0.002	+0.005	+0.015	-0.038	-0.017	+0.018
Finnish (363)	-0.024	+0.022	+0.181	+0.096	+0.052	-0.024	+0.012	-0.013
Italian (201)	+0.030	-0.002	+0.111	+0.058	+0.019	+0.047	-0.011	-0.004
German (191)	+0.046	+0.154	+0.104	+0.169	+0.073	+0.028	+0.022	+0.046
Swedish (97)	+0.005	+0.116	+0.017	+0.010	+0.024	+0.012	-0.001	+0.017
Norwegian (83)	+0.018	+0.084	-0.009	-0.004	+0.009	+0.013	+0.013	+0.012
French (83)	+0.230	+0.064	+0.016	+0.004	-0.173	+0.054	-0.019	-0.101
Dutch (56)	+0.011	-0.044	-0.020	-0.007	+0.072	+0.072	+0.018	+0.053

Table 15: Per-language translated–native differences in Task B macro-F1. Values are $\Delta = F1_{\text{translated}} - F1_{\text{native}}$, so positive values indicate higher performance on English-translated input. 0s and 5s denote zero-shot and five-shot prompting. Results are shown for languages with at least 50 posts in the gold set.

Model	Text	Setting	Task A	Task B	Task B.1
Gemma 4	translated	0-shot	0.658 [0.634, 0.682]	0.352 [0.319, 0.388]	0.563 [0.522, 0.605]
Gemma 4	translated	5-shot	0.624 [0.597, 0.650]	0.331 [0.293, 0.369]	0.553 [0.505, 0.602]
Gemma 4	native	0-shot	0.633 [0.606, 0.658]	0.346 [0.314, 0.380]	0.552 [0.519, 0.585]
Gemma 4	native	5-shot	0.566 [0.536, 0.593]	0.292 [0.252, 0.332]	0.497 [0.459, 0.537]
GPT-4.1-mini	translated	0-shot	0.638 [0.609, 0.666]	0.359 [0.320, 0.399]	0.494 [0.472, 0.515]
GPT-4.1-mini	translated	5-shot	0.617 [0.586, 0.646]	0.325 [0.289, 0.362]	0.466 [0.443, 0.488]
GPT-4.1-mini	native	0-shot	0.616 [0.588, 0.644]	0.328 [0.293, 0.363]	0.491 [0.469, 0.512]
GPT-4.1-mini	native	5-shot	0.596 [0.566, 0.626]	0.315 [0.281, 0.351]	0.463 [0.441, 0.484]
GPT-4o-mini	translated	0-shot	0.620 [0.593, 0.646]	0.392 [0.346, 0.437]	0.646 [0.591, 0.698]
GPT-4o-mini	translated	5-shot	0.624 [0.595, 0.651]	0.381 [0.331, 0.429]	0.660 [0.601, 0.714]
GPT-4o-mini	native	0-shot	0.601 [0.573, 0.627]	0.384 [0.338, 0.428]	0.623 [0.565, 0.677]
GPT-4o-mini	native	5-shot	0.605 [0.576, 0.633]	0.372 [0.326, 0.418]	0.650 [0.603, 0.697]
Qwen 3.5	translated	0-shot	0.567 [0.538, 0.595]	0.277 [0.244, 0.313]	0.472 [0.448, 0.496]
Qwen 3.5	translated	5-shot	0.584 [0.554, 0.612]	0.269 [0.235, 0.307]	0.441 [0.413, 0.474]
Qwen 3.5	native	0-shot	0.558 [0.528, 0.587]	0.277 [0.242, 0.315]	0.461 [0.438, 0.486]
Qwen 3.5	native	5-shot	0.573 [0.543, 0.601]	0.242 [0.209, 0.277]	0.430 [0.407, 0.458]

Table 16: LLM macro-F1 with 95% percentile bootstrap confidence intervals from 10,000 stratified-bootstrap replicates. Invalid predictions are retained and scored as incorrect.

Model	Setting	Task A	Task B	Task B.1
Gemma 4	0-shot	+0.025 [+0.007, +0.045]	+0.007 [-0.014, +0.026]	+0.012 [-0.017, +0.038]
Gemma 4	5-shot	+0.058 [+0.033, +0.085]	+0.039 [+0.004, +0.076]	+0.056 [+0.019, +0.092]
GPT-4.1-mini	0-shot	+0.022 [+0.002, +0.043]	+0.032 [+0.010, +0.054]	+0.003 [-0.015, +0.021]
GPT-4.1-mini	5-shot	+0.021 [-0.002, +0.043]	+0.009 [-0.013, +0.033]	+0.003 [-0.016, +0.021]
GPT-4o-mini	0-shot	+0.019 [-0.001, +0.039]	+0.008 [-0.025, +0.039]	+0.023 [-0.004, +0.058]
GPT-4o-mini	5-shot	+0.018 [-0.002, +0.038]	+0.009 [-0.022, +0.037]	+0.010 [-0.034, +0.049]
Qwen 3.5	0-shot	+0.009 [-0.016, +0.035]	+0.000 [-0.019, +0.020]	+0.010 [-0.013, +0.035]
Qwen 3.5	5-shot	+0.011 [-0.012, +0.034]	+0.027 [-0.001, +0.052]	+0.011 [-0.009, +0.036]

Table 17: Paired translated–native differences in LLM macro-F1. Values are $\Delta = F1_{\text{translated}} - F1_{\text{native}}$ with 95% percentile bootstrap confidence intervals from 10,000 stratified-bootstrap replicates. Positive values favor translated input. Intervals excluding zero indicate evidence for a difference under this bootstrap analysis.

H.5 Classwise performance

Figures 4-9 show substantial variation across classes, while Table 18 gives the key used for each label. On Task A, most systems perform particularly strongly on service disruption and infrastructure posts, while water-quality posts remain more difficult. On Task B, performance is considerably lower for rare classes such as operational resolution, reassurance, and other or unclear communication. Grouping the overlapping communication functions in Task B.1 produces more stable classwise performance, particularly for risk and incident communication.

H.6 Detailed classwise results

Tables 19-24 provide precision, recall, F1, and support for the strongest supervised baseline and strongest LLM configuration on each task, complementing the classwise visualizations above.

Task	Code	Class label
Task A	A1	Water quality / safety / public health
Task A	A2	Service disruption / repair / infrastructure
Task A	A3	Water conservation / drought / demand management
Task A	A4	Flood / stormwater / wastewater / sewer
Task A	A5	Environmental sustainability / waste / recycling / biodiversity
Task A	A6	Public education / heritage / community engagement
Task A	A7	Routine / institutional / customer service / other
Task B	B1	Alert / warning
Task B	B2	Instruction / advice to public
Task B	B3	Operational update / resolution
Task B	B4	Reassurance / safety information
Task B	B5	Education / awareness
Task B	B6	Institutional promotion / community news
Task B	B7	Other / unclear
Task B.1	G1	Risk / incident communication
Task B.1	G2	Education / awareness
Task B.1	G3	Institutional / community communication
Task B.1	G4	Other / unclear

Table 18: Class codes used in the classwise evaluation figures and supplementary tables.

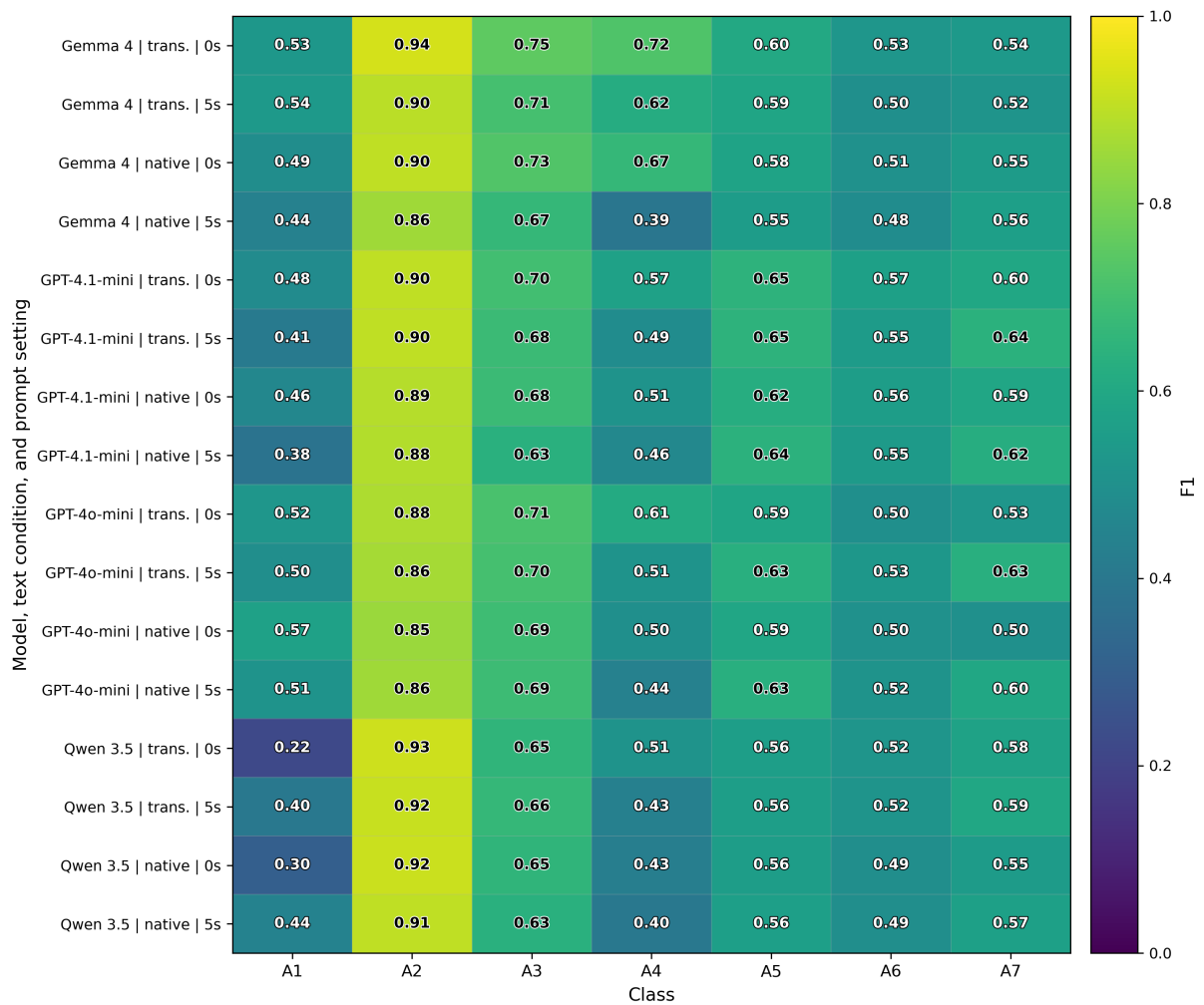


Figure 4: Classwise F1 for LLMs on Task A. Values are computed on all 1,500 gold-standard posts. A1-A7 denote the Task A labels defined in Table 18.



Figure 5: Classwise F1 for LLMs on Task B. Values are computed on all 1,500 gold-standard posts. B1-B7 denote the Task B labels defined in Table 18.

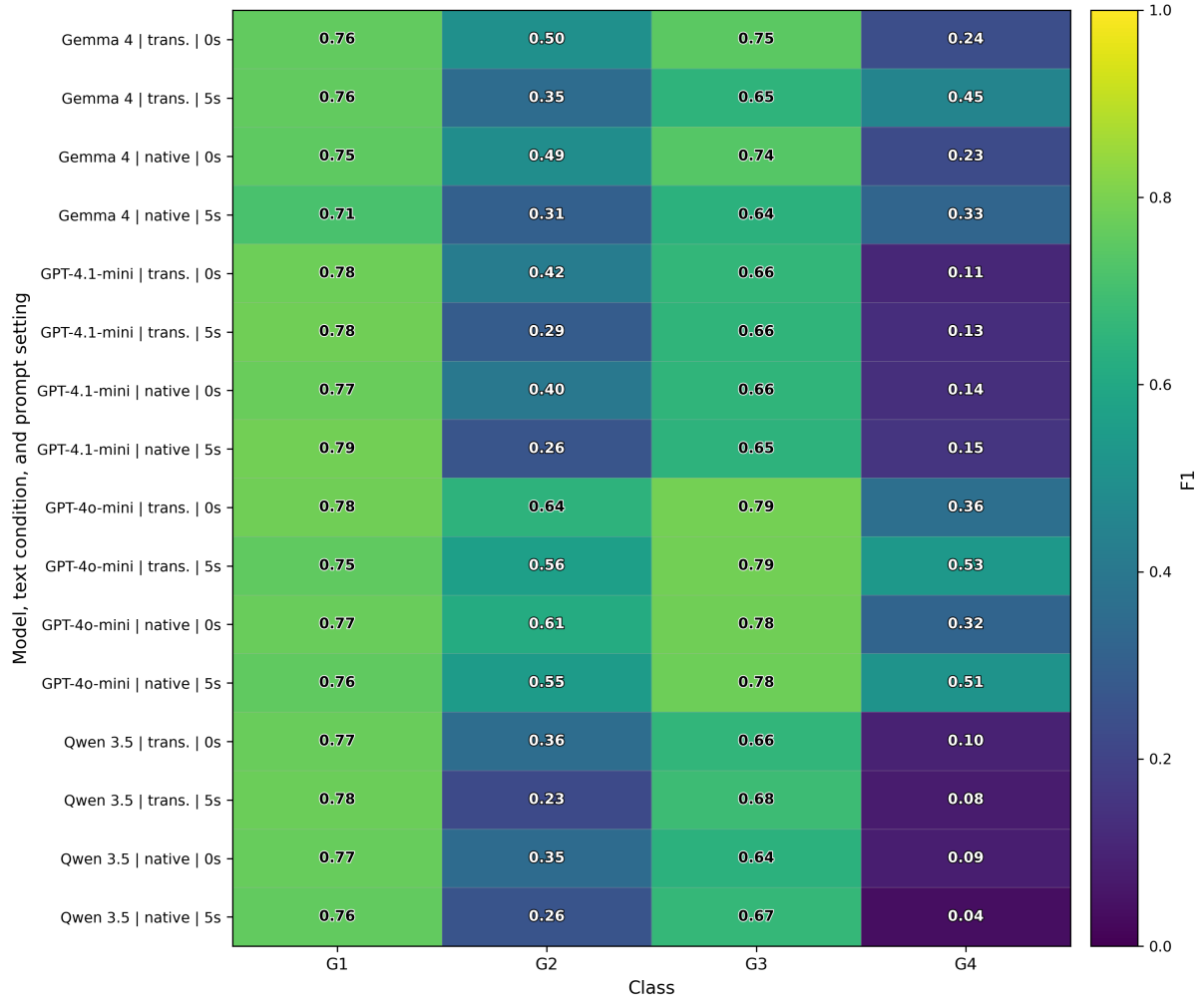


Figure 6: Classwise F1 for LLMs on grouped Task B.1. Values are computed on all 1,500 gold-standard posts. G1-G4 denote the grouped communication-function labels defined in Table 18.

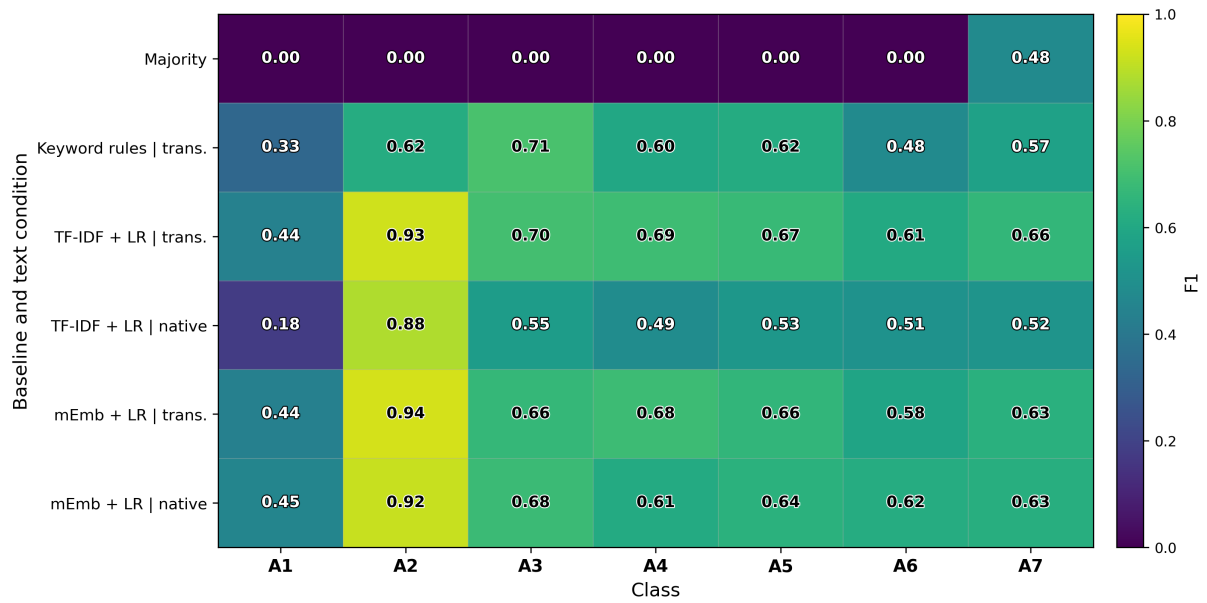


Figure 7: Classwise F1 for all baselines on Task A. Values are computed from pooled out-of-fold predictions under stratified five-fold cross-validation. A1-A7 denote the Task A labels defined in Table 18.

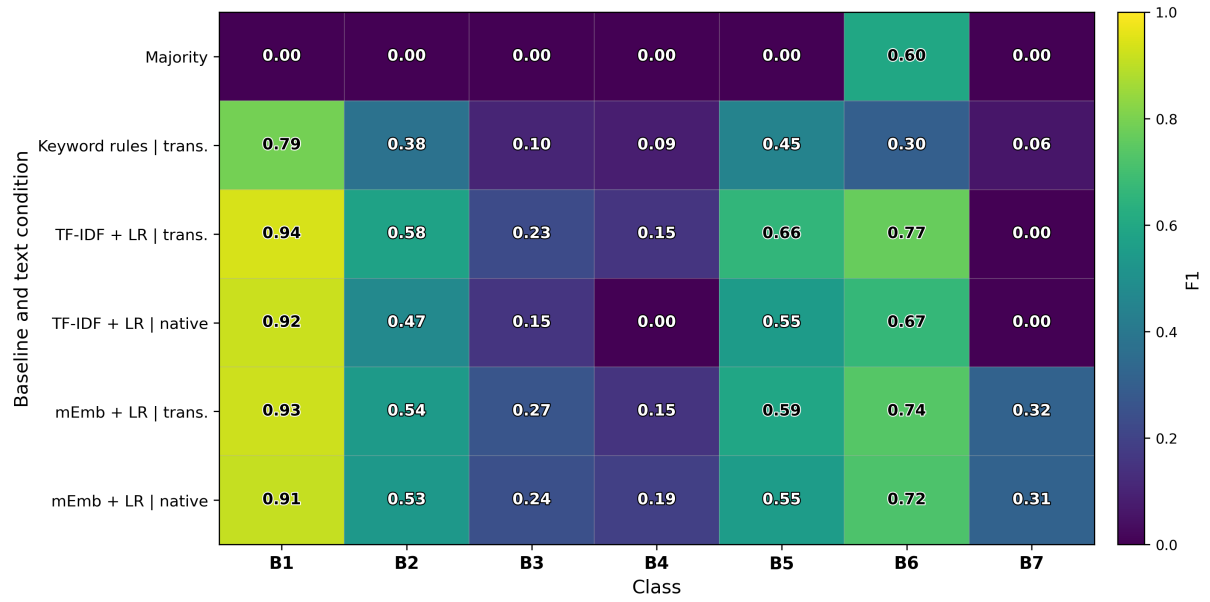


Figure 8: Classwise F1 for all baselines on Task B. Values are computed from pooled out-of-fold predictions under stratified five-fold cross-validation. B1-B7 denote the Task B labels defined in Table 18.

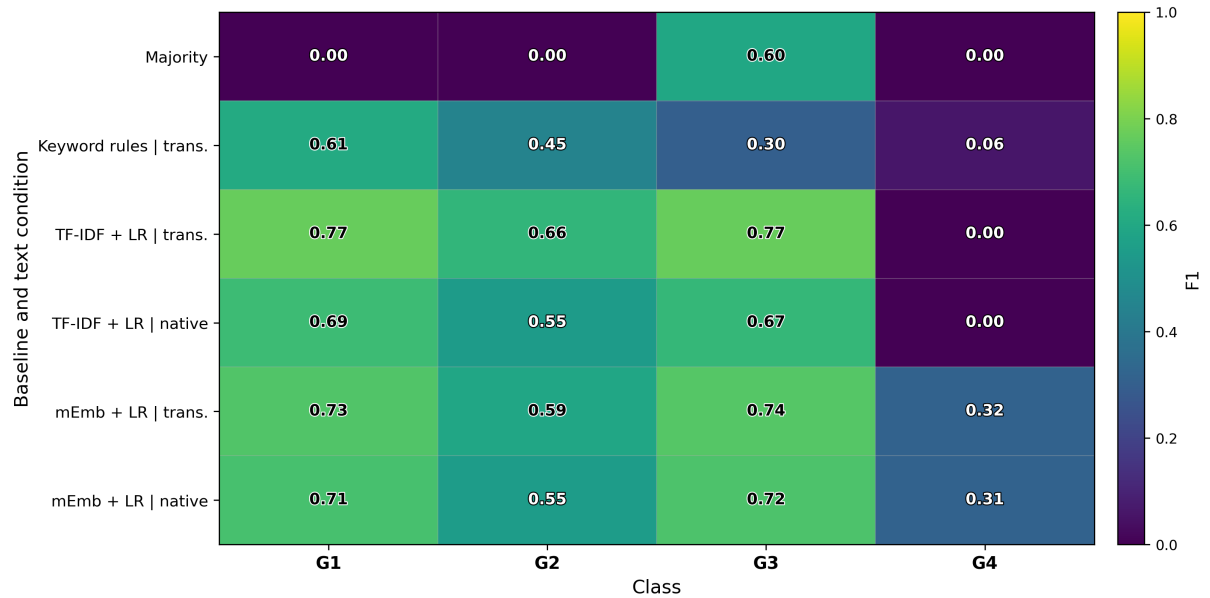


Figure 9: Classwise F1 for baselines on grouped Task B.1. Values are computed from pooled out-of-fold predictions under stratified five-fold cross-validation. G1-G4 denote the grouped communication-function labels defined in Table 18.

Model	Text	Setting	Class	Precision	Recall	F1	Support
Gemma 4	translated	zero-shot	A1	0.481	0.585	0.528	65
Gemma 4	translated	zero-shot	A2	0.923	0.951	0.937	226
Gemma 4	translated	zero-shot	A3	0.738	0.764	0.751	140
Gemma 4	translated	zero-shot	A4	0.714	0.735	0.725	136
Gemma 4	translated	zero-shot	A5	0.469	0.842	0.603	209
Gemma 4	translated	zero-shot	A6	0.467	0.604	0.527	250
Gemma 4	translated	zero-shot	A7	0.888	0.384	0.536	474

Table 19: Classwise precision, recall, and F1 for the strongest LLM on Task A: Gemma 4, translated, zero-shot. Support is the number of gold examples in each class.

Model	Text	Setting	Class	Precision	Recall	F1	Support
GPT-4o-mini	translated	zero-shot	B1	0.706	0.059	0.108	205
GPT-4o-mini	translated	zero-shot	B2	0.442	0.530	0.482	164
GPT-4o-mini	translated	zero-shot	B3	0.069	0.667	0.124	21
GPT-4o-mini	translated	zero-shot	B4	0.200	0.273	0.231	11
GPT-4o-mini	translated	zero-shot	B5	0.671	0.621	0.645	446
GPT-4o-mini	translated	zero-shot	B6	0.791	0.788	0.790	638
GPT-4o-mini	translated	zero-shot	B7	0.333	0.400	0.364	15

Table 20: Classwise precision, recall, and F1 for the strongest LLM on Task B: GPT-4o-mini, translated, zero-shot. Support is the number of gold examples in each class.

Model	Text	Setting	Class	Precision	Recall	F1	Support
GPT-4o-mini	translated	five-shot	G1	0.875	0.663	0.755	401
GPT-4o-mini	translated	five-shot	G2	0.732	0.460	0.565	446
GPT-4o-mini	translated	five-shot	G3	0.673	0.950	0.788	638
GPT-4o-mini	translated	five-shot	G4	0.533	0.533	0.533	15

Table 21: Classwise precision, recall, and F1 for the strongest LLM on Task B.1: GPT-4o-mini, translated, five-shot. Support is the number of gold examples in each class.

Model	Text	Setting	Class	Precision	Recall	F1	Support
TF-IDF + LR	translated	Stratified 5-fold CV	A1	0.378	0.523	0.439	65
TF-IDF + LR	translated	Stratified 5-fold CV	A2	0.990	0.872	0.927	226
TF-IDF + LR	translated	Stratified 5-fold CV	A3	0.685	0.714	0.699	140
TF-IDF + LR	translated	Stratified 5-fold CV	A4	0.598	0.809	0.688	136
TF-IDF + LR	translated	Stratified 5-fold CV	A5	0.703	0.646	0.673	209
TF-IDF + LR	translated	Stratified 5-fold CV	A6	0.542	0.692	0.608	250
TF-IDF + LR	translated	Stratified 5-fold CV	A7	0.754	0.589	0.661	474

Table 22: Classwise precision, recall, and F1 for the strongest supervised baseline on Task A: TF-IDF + LR using translated text. Scores are computed from pooled out-of-fold predictions under stratified five-fold cross-validation. Support is the number of gold examples in each class.

Model	Text	Setting	Class	Precision	Recall	F1	Support
Multilingual embeddings + LR	translated	Stratified 5-fold CV	B1	0.944	0.907	0.925	205
Multilingual embeddings + LR	translated	Stratified 5-fold CV	B2	0.436	0.726	0.545	164
Multilingual embeddings + LR	translated	Stratified 5-fold CV	B3	0.177	0.524	0.265	21
Multilingual embeddings + LR	translated	Stratified 5-fold CV	B4	0.103	0.273	0.150	11
Multilingual embeddings + LR	translated	Stratified 5-fold CV	B5	0.647	0.543	0.590	446
Multilingual embeddings + LR	translated	Stratified 5-fold CV	B6	0.816	0.669	0.736	638
Multilingual embeddings + LR	translated	Stratified 5-fold CV	B7	0.214	0.600	0.316	15

Table 23: Classwise precision, recall, and F1 for the strongest supervised baseline on Task B: multilingual embeddings + LR using translated text. Scores are computed from pooled out-of-fold predictions under stratified five-fold cross-validation. Support is the number of gold examples in each class.

Model	Text	Setting	Class	Precision	Recall	F1	Support
Multilingual embeddings + LR	translated	Stratified 5-fold CV	G1	0.624	0.873	0.728	401
Multilingual embeddings + LR	translated	Stratified 5-fold CV	G2	0.647	0.543	0.590	446
Multilingual embeddings + LR	translated	Stratified 5-fold CV	G3	0.816	0.669	0.736	638
Multilingual embeddings + LR	translated	Stratified 5-fold CV	G4	0.214	0.600	0.316	15

Table 24: Classwise precision, recall, and F1 for the strongest supervised baseline on grouped Task B.1: multilingual embeddings + LR using translated text. Scores are computed from pooled out-of-fold predictions under stratified five-fold cross-validation. Support is the number of gold examples in each class.